SMARTMAINT AI: AI BASED PREDICTIVE MAINTENANCE SYSTEM FOR ROTATING EQUIPMENT

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SMARTMAINT AI: AI BASED PREDICTIVE MAINTENANCE SYSTEM FOR ROTATING EQUIPMENT

ALWALEED HASSAN ELYAS HASSAN

This report is submitted in partial fulfillment of the requirements for the Bachelor of [Computer Science (Artificial Intelligence)] with Honours.

FACULTY OF ARTIFICIAL INTELLIGENCE AND CYBER SECURITY
UNIVERSITI TEKNIKAL MALAYSIA MELAKA

2026

DECLARATION

I hereby declare that this project report entitled

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SUPERVISOR : ________________________ Date : 29 June, 2026
(Azah Kamilah Draman Binti Muda)

DECLARATION

DEDICATION

I dedicate this project to my beloved parents and family for their endless love, support, prayers, and encouragement throughout my academic journey. Their patience, sacrifices, and belief in me gave me the strength to continue working hard and complete this Final Year Project.
Their support has always been a source of motivation, especially during difficult times. I am truly grateful for everything they have done for me, and I dedicate this achievement to them with love and appreciation.

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ABSTRACT

The project is in the field of artificial intelligence and predictive maintenance, focusing on rotating equipment such as motors, pumps, fans, compressors, and other industrial machines. Unexpected machine failures can cause operational downtime, high repair costs, and reduced productivity. This project develops SmartMaint AI, a local PC-based predictive maintenance application that analyses machine operation data and supports early maintenance decision-making.

The system uses the AI4I 2020 Predictive Maintenance dataset. Random Forest is implemented as the main supervised machine-failure prediction model because it achieved the strongest evaluation performance. Isolation Forest is retained as a supporting unsupervised anomaly-detection model, while One-Class SVM is used for unsupervised comparison. The system performs data preprocessing, model training and evaluation, machine-failure prediction, anomaly detection, simulated fleet health assessment, alert-level classification, maintenance recommendation, alert-log generation, dashboard visualisation, and report generation.

Because the AI4I dataset does not contain real fleet identities or equipment histories, SmartMaint AI creates a controlled simulated fleet of 200 machines across 18 equipment categories for prototype demonstration. The application was developed using Python and Streamlit. Users can monitor machine conditions, review prediction evidence, check maintenance alerts, inspect recommended actions, obtain maintenance schedule suggestions, and export reports. The completed system demonstrates how machine learning and decision-support logic can be integrated into a practical predictive-maintenance product.

ABSTRAK

Projek ini adalah dalam bidang kecerdasan buatan dan penyelenggaraan ramalan, dengan tumpuan kepada peralatan berputar seperti motor, pam, kipas, pemampat dan mesin perindustrian lain. Kegagalan mesin yang tidak dijangka boleh menyebabkan masa henti operasi, kos pembaikan yang tinggi dan produktiviti yang berkurangan. Projek ini membangunkan SmartMaint AI, aplikasi penyelenggaraan ramalan berasaskan PC tempatan yang menganalisis data operasi mesin dan menyokong pembuatan keputusan penyelenggaraan awal.

Sistem ini menggunakan set data Penyelenggaraan Ramalan AI4I 2020. Hutan Rawak dilaksanakan sebagai model ramalan kegagalan mesin yang diselia utama kerana ia mencapai prestasi penilaian yang paling kuat. Hutan Pengasingan dikekalkan sebagai model pengesanan anomali tanpa penyeliaan yang menyokong, manakala SVM Satu Kelas digunakan untuk perbandingan tanpa penyeliaan. Sistem ini melakukan prapemprosesan data, latihan dan penilaian model, ramalan kegagalan mesin, pengesanan anomali, penilaian kesihatan armada simulasi, pengelasan tahap amaran, cadangan penyelenggaraan, penjanaan log amaran, visualisasi papan pemuka dan penjanaan laporan.

Oleh kerana set data AI4I tidak mengandungi identiti armada sebenar atau sejarah peralatan, SmartMaint AI mencipta armada simulasi terkawal yang terdiri daripada 200 mesin merentasi 18 kategori peralatan untuk demonstrasi prototaip. Aplikasi ini dibangunkan menggunakan Python dan Streamlit. Pengguna boleh memantau keadaan mesin, menyemak bukti ramalan, menyemak amaran penyelenggaraan, memeriksa tindakan yang disyorkan, mendapatkan cadangan jadual penyelenggaraan dan mengeksport laporan. Sistem yang telah siap ini menunjukkan bagaimana pembelajaran mesin dan logik sokongan keputusan boleh disepadukan ke dalam produk penyelenggaraan ramalan yang praktikal.

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FYP	**-**	**Final Year Project**

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CHAPTER 1: INTRODUCTION

1.1 Introduction

Predictive maintenance is an important application of artificial intelligence that helps identify possible machine problems before serious breakdowns occur. Rotating equipment such as motors, pumps, fans, compressors, generators, and gearboxes is widely used in many industries. A sudden machine failure may increase repair costs, interrupt production, reduce equipment availability, and create safety risks. Therefore, maintenance should not depend only on manual inspection or waiting until a machine breaks down.

This project develops SmartMaint AI, a local PC-based predictive maintenance system for rotating equipment. The system uses machine operating data and machine learning techniques to predict machine failure, identify unusual operating patterns, and present maintenance information through an interactive Streamlit dashboard.

Random Forest is used as the main supervised machine failure prediction model because it achieved the best overall evaluation performance. Isolation Forest is used as a supporting unsupervised anomaly detection model, while One-Class SVM is used as an unsupervised comparison model.

SmartMaint AI is developed as a product-based Final Year Project. The final product is a working application that displays machine health scores, alert levels, maintenance priorities, detected problems, probable causes, observed abnormal patterns, recommended maintenance actions, maintenance alerts, schedule suggestions, and downloadable reports.

The system is designed to make machine condition information easier to understand. It supports users in identifying machines that may require observation, inspection, urgent maintenance, or immediate attention before the condition becomes more serious.

1.2 Problem Statement

In most maintenance activities, machine problems are detected only after the machine exhibits clearly visible defects or stops operating. This reactive approach often produces unexpected downtime, high repair cost, and loss of productivity. It creates problems for maintenance teams also since urgent repairs usually take more time, cost, and manpower. Also, there is a human factor in manual monitoring of machines. Different technicians may perceive the same machine condition differently. This can lead to late detection, inconsistent judgment, or missed early warning signs. The fact that there are many machines to monitor makes it difficult to check all machines manually and continuously. Machine learning can help to detect abnormal machine behavior, but it must be presented in a simple and understandable manner. A prediction result alone may not be helpful if the user does not know what action should be taken. This means that there is a need for a system to analyze machine data, classify the machine condition, and give maintenance information (health score, alert level, problem description, possible cause, and recommended action).

1.3 Objectives

(a) To develop a local PC-based predictive maintenance application for rotating equipment using Python and Streamlit.

(b) To implement Random Forest as the main supervised machine-learning technique for machine-failure prediction.

(c) To apply Isolation Forest as a supporting anomaly-detection technique and One-Class SVM as an unsupervised comparison model.

(d) To generate useful maintenance outputs, including machine health scores, alert levels, detected problems, probable causes, observed abnormal patterns, recommended actions, maintenance schedules, maintenance alert logs, and report summaries.

1.4 Scope

The scope of this project focuses on developing a predictive maintenance application for rotating equipment using the AI4I 2020 Predictive Maintenance dataset. The system processes machine operating data, including air temperature, process temperature, rotational speed, torque, tool wear, and machine type, to predict machine failure and identify unusual operating patterns.

Random Forest is implemented as the main supervised machine learning model for machine failure prediction. Isolation Forest is used as a supporting unsupervised anomaly detection model, while One-Class SVM is used as an unsupervised comparison model.

The system also generates a controlled simulated fleet of 200 machines from 18 rotating-equipment categories for prototype demonstration. The simulated machine conditions are represented using health scores, alert levels, and maintenance priorities.

SmartMaint AI provides maintenance-related outputs such as detected problems, probable causes, observed abnormal patterns, recommended actions, maintenance schedule suggestions, maintenance alerts, and downloadable reports. These outputs are displayed through a local Streamlit dashboard.

The project is limited to a local PC-based application using stored dataset records. It does not include live sensor connections, cloud deployment, or direct integration with industrial machines. The health scores and maintenance recommendations are used for prototype demonstration and decision support only.

1.4.1 Target User

The target users of this system are maintenance staff, technicians, engineers, and students who need to monitor machine condition and understand predictive maintenance results. The system is designed to be simple and clear so that users can identify which machines are healthy and which machines require attention.

1.4.2 Modules

a) **Dashboard Module**
This module provides an overview of the whole system, including total machines, health condition summary, alert distribution, and recent high-risk machines.

b) **Machine Monitoring Module**
This module allows users to view machine details such as machine ID, equipment category, health score, alert level, detected problem, probable cause, abnormal pattern, and recommended action.

c) **AI Prediction and Health Scoring Module**
This module applies machine learning techniques to analyse machine data, detect abnormal behaviour, and calculate the machine health score.

d) **Maintenance Alert Log Module**
This module displays active maintenance alerts generated from the AI results. It helps users identify machines that need inspection, urgent action, or immediate maintenance.

e) **Report Generation Module**
This module allows users to export machine information and maintenance summaries for documentation and presentation purposes.

1.5 Project Significance

The project is important as it illustrates the application of artificial intelligence to a practical maintenance system. Instead of just manually checking the machine, it helps users identify abnormal machine behavior earlier. This can lead to better maintenance planning and less chance of unexpected machine failure. Machine learning results can be visualized in the system, and not only does it present prediction output, but also health score, alert level, detected problem, probable cause and recommended action. This makes the system even more useful for non-technical users who don't have a deep technical knowledge of machine learning. Besides that, the maintenance alert log makes the project more realistic as a product-based system. It turns AI results into maintenance information that can help inform decision making. As such, this project can help us to demonstrate the importance of machine learning in predictive maintenance and to demonstrate how AI can be used to solve real industrial problems.

1.6 Conclusion

In conclusion, in this chapter i have highlighted the background, problem statement, objectives, scope, target users, modules, and significance of the SmartMaint AI project. The project focuses on developing a local PC-based predictive maintenance application that utilizes machine learning to detect abnormal machine behavior and support maintenance decisions. The next chapter will discuss the literature review and project methodology with regard to predictive maintenance, anomaly detection, as well as machine learning in this project.

CHAPTER 2: LITERATURE REVIEW AND PROJECT METHODOLOGY

2.1 Introduction

This chapter explains the literature review and project methodology used for the SmartMaint AI project. The purpose of this chapter is to give basic background about predictive maintenance, rotating equipment, machine learning techniques, the selected dataset, and the development method used in this project.

Rotating equipment such as motors, pumps, fans, compressors, generators, and gearboxes is commonly used in many industries. If one of these machines suddenly fails, it can cause machine downtime, high repair costs, production delays, and safety problems. Because of this, maintenance should not only depend on manual inspection or a fixed maintenance schedule. A data-based maintenance system can help detect machine problems earlier before they become more serious.

Machine learning can be used in predictive maintenance to analyse machine operating data and detect unusual patterns. In this project, the system uses data such as air temperature, process temperature, rotational speed, torque, tool wear, and machine type. These features help the system understand the machine condition and identify possible failure risks or abnormal behaviour.

SmartMaint AI uses Random Forest as the main supervised machine learning model for machine failure prediction because it achieved the best evaluation results. Isolation Forest is used as a supporting anomaly detection model, while One-Class SVM is used as an unsupervised comparison model. These three models have different roles in the system and help compare supervised prediction with unsupervised anomaly detection.

The AI4I 2020 Predictive Maintenance dataset is used as the main dataset in this project. However, the dataset does not provide real machine IDs, equipment histories, or complete fleet information. Therefore, the system creates a controlled simulated fleet for dashboard and system demonstration. The original sensor readings and machine learning outputs are still used as AI prediction evidence, while the simulated machine health conditions are used to show health scores, alert levels, maintenance recommendations, alert logs, and reports.

SmartMaint AI is developed using the Waterfall Model. The main stages are requirement gathering and analysis, system design, implementation, testing and evaluation, and documentation. Overall, this chapter gives the main background needed to understand the project and explains how machine learning, predictive maintenance, the dataset, and the development methodology are used to build SmartMaint AI as a product-based system.

2.2 Facts and Findings

This section presents the main facts and findings related to predictive maintenance, rotating equipment, machine learning, and the dataset used in this project. These findings are important because they help explain the problem and support the development of SmartMaint AI as a predictive maintenance application.

The main challenge in this project is not only to predict machine failure, but also to help users understand the machine condition before the problem becomes serious. A simple prediction result may not be enough for maintenance users. They also need clear information such as machine health score, alert level, detected problem, possible cause, recommended action, and maintenance priority.

This section explains how machine operating data can be used to identify abnormal behaviour and possible failure risk. It also discusses how machine learning can support maintenance decisions and how the results can be presented in a simple and useful form through health scores, alert levels, maintenance recommendations, alert logs, and reports.

Random Forest is used as the main supervised machine learning model for machine failure prediction because it achieved the best evaluation performance. Isolation Forest is used as a supporting anomaly detection model because it can identify unusual machine operating patterns without using failure labels. One-Class SVM is also included as an unsupervised comparison model.

This section also discusses the predictive maintenance domain, existing maintenance methods, the selected machine learning techniques, and the AI4I 2020 Predictive Maintenance dataset. Overall, it provides the background needed to understand how SmartMaint AI combines machine learning with a practical maintenance decision-support system.

2.2.1 Domain

Predictive maintenance in artificial intelligence is the field of this project. Predictive maintenance is a method of maintenance that is concerned with identifying machine problems and preventing them from becoming critical failures. Predictive maintenance is not waiting for a machine to break down but monitors the machine to monitor its state and identify early warning signs. There is a lot of rotating equipment in the industry like motors, pumps, fans, compressors, gearboxes, and generators. These machines are essential as they work in much of the industry. If one machine fails suddenly, it may cause operation delay, high repair cost, production downtime, and safety issues. Therefore, monitoring the condition of rotating equipment is very important.

Old methods of maintenance usually rely on manual inspection or fixed maintenance schedules. However, manual inspection may be affected by human error and fixed schedules may not always be right for the machine. Some machines may still be healthy but are serviced too early, while other machines may fail well before the scheduled maintenance date. Predictive maintenance is a better option as data is used to make maintenance decisions. Machine learning is used in this project for machine operating data analysis and to detect abnormal behavior. The system uses machine features like air temperature, process temperature, rotational speed, torque, and tool wear. This allows the machine to understand the machine's condition and classify the risk level. The final output is made available on a dashboard of health score, alert level, detected problem, probable cause, recommended action, maintenance alert log.

2.2.1.1 Predictive Maintenance

Predictive maintenance is a maintenance procedure for identifying potential machine problems before the machine fails. It is based on machine data to monitor current machine state and spot early signs of abnormal behavior. This is different from reactive maintenance which takes action only after the machine has already failed. It is also different from preventive maintenance, which is carried out according to a fixed schedule even though the machine is still in good condition. The benefits of predictive maintenance are that it minimizes unexpected downtime. If abnormal machine patterns are detected earlier, maintenance teams can plan inspections or repairs before the machine is at its worst. This can save money in repairs, increase reliability, avoid a sudden production interruption, and improve maintenance planning. For industrial equipment that runs continuously, early detection is crucial because even a small problem can be a serious failure if not detected.

In this project, predictive maintenance is conducted with machine learning models to analyze machine operation data. The system processes machine temperature, process temperature, rotational speed, torque, and tool wear and aims to detect abnormal machine behaviors. The results from machine learning are then converted to practical maintenance information like machine health score, alert level, detected problem, probable cause, and recommended action.

This makes the system easier to understand as the user is not only receiving a technical prediction result, but also useful information that can help in maintenance decision making.

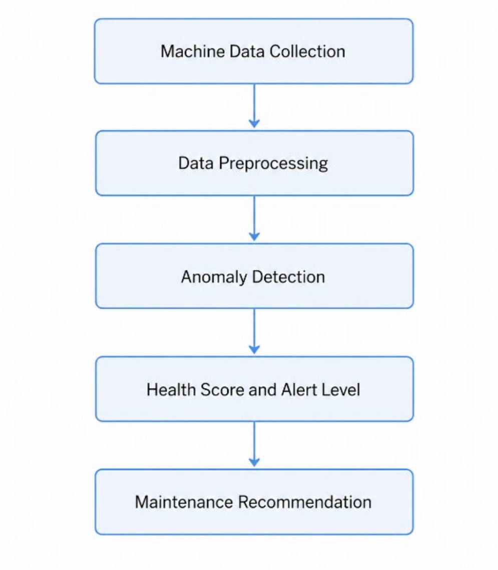


Figure 2.1: Predictive Maintenance Process flow

2.2.1.2 Machine Learning in Predictive Maintenance

Machine learning is useful for predictive maintenance because it can learn patterns from machine operating data and use these patterns to identify possible failures or abnormal behaviour. During normal operation, machine sensor readings usually follow a certain pattern. When a machine starts to degrade or operate under an unusual condition, changes may appear in readings such as temperature, rotational speed, torque, and tool wear.

These changes may be difficult to notice manually, especially when many machines need to be monitored at the same time. Machine learning can analyse a large amount of data and help users identify machines that may require inspection or maintenance. This allows maintenance decisions to be based on machine data instead of depending only on manual inspection or fixed maintenance schedules.

SmartMaint AI uses three machine learning models, but each model has a different role. Random Forest is used as the main supervised machine learning model for machine failure prediction. It learns from the machine operating features and the labelled machine failure data. Random Forest was selected as the main model because it achieved the best overall performance during model evaluation.

Isolation Forest is used as a supporting anomaly detection model. It identifies machine records that are different from the normal operating pattern without directly using the machine failure label. This is useful because abnormal and failure records are less common than normal operating records.

One-Class SVM is also used as an unsupervised anomaly detection model for comparison. It learns the general boundary of normal machine behaviour and identifies records that fall outside this boundary. Comparing One-Class SVM with Isolation Forest helps the project understand how different unsupervised models detect unusual machine conditions.

Using these three models makes the project stronger because the system can compare supervised failure prediction with unsupervised anomaly detection. Random Forest provides the main failure prediction, Isolation Forest provides supporting anomaly evidence, and One-Class SVM is used as a comparison model.

2.2.1.3 Dataset

The project will utilize the AI4I 2020 Predictive Maintenance dataset for developing and testing the SmartMaint AI system. The dataset contains machine operations data for predictive maintenance studies and there are a number of key features such as air temperature, process temperature, rotational speed, torque, tool wear and machine failure information. The dataset should be readily useful in this work as the features are typical operating conditions of industrial machines. For instance, the temperature can be used to determine whether the machine is operating in normal or abnormal heat conditions. Rotational speed and torque can also show the machine’s mechanical behaviour during operation. Tool wear can indicate how much the machine tool has been used and if degradation will happen in operation. These features are useful because machine failure is often due to abnormal changes in operating conditions. SmartMaint AI uses the data in several stages. First, i load and preprocess the data and the data is then trained for machine learning. Then, the selected machine learning models are used to identify abnormal patterns and compare prediction performance.

The model output is then converted to more understandable maintenance information like machine health score, alert level, detected problem, probable cause, and recommended action. This dataset also enables the project to demonstrate predictive maintenance without the need for real industrial sensors yet. With this kind of project as a local PC-based prototype, it is safe to use an existing predictive maintenance dataset for training, testing and demonstrating the system. In the future, it should be better to connect it to real-time sensor data from actual rotating equipment. The use of the AI4I 2020 Predictive Maintenance dataset supports the main goal of this project. This enables SmartMaint AI to analyse machine conditions, detect abnormal behaviour, classify risk levels and provide maintenance decision support in a practical manner.

Table 2.1: Predictive Maintenance Process flow

Feature	Description	Use in SmartMaint AI
Air Temperature	Surrounding air temperature of the machine	Used to monitor heat condition
Process Temperature	Temperature during machine operation	Used to detect abnormal process heat
Rotational Speed	Machine rotation speed in rpm	Used to identify abnormal speed pattern
Torque	Load or force during machine operation	Used to detect mechanical stress
Tool Wear	Tool usage or wear level	Used to estimate degradation
Machine Failure	Failure status of the machine	Used for comparison and evaluation

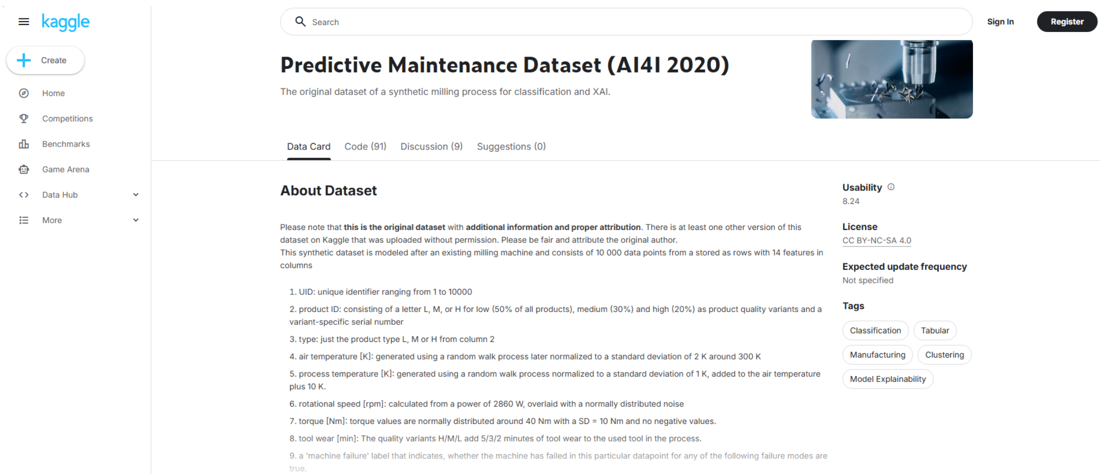


Figure 2.2: Kaggule dataset

2.2.2 Existing System

Current maintenance systems generally have a reactive maintenance method and preventive maintenance. If a machine suddenly stops working, the maintenance team will have to act fast, and this way pressure, cost and repair time will be increased. Preventive maintenance is a common method in which machines are serviced according to a timetable. For example, a machine would be inspected on a weekly basis, monthly or after a certain number of operating hours. This is a better option than waiting for failure but still has some limitations.

Some machines will be serviced despite being in good condition and some machines will fail before the scheduled maintenance date. That means preventive maintenance does not always reflect the actual state of the machine. Modern systems use sensor information and machine learning to predict the system's conditions. These systems can identify machine behavior and could also be able to detect failure patterns or patterns in a machine.

However, most current systems only analyze the prediction results of future problems and do not explain which parameters to use. So, for example, the system may tell you if a machine is normal or not, but the machine may not give you the cause of the problem, alert level or recommended maintenance action. Some of them will need hardware, real time sensors, IoT devices or cloud platforms to be able to do that. If the system is expensive or difficult to use for students and small projects or early prototype development, for example, the system may be expensive and difficult to use. So, it is necessary to have a simple predictive maintenance application which can run locally on the PC and still offer useful maintenance information. The proposed SmartMaint AI system gives this better functionality as it is a local PC based predictive maintenance application. It not only detects the abnormal behavior of the machine but also presents the result more clearly. It provides machine health score, alert level, detected problem, probable cause, recommended action, maintenance alert log, and report generation. This makes the system more real for the users as the prediction result is converted into maintenance decision support.

2.2.3 Proposed Technique

The proposed technique for this project is a machine learning-based predictive maintenance approach. The main purpose of this approach is to analyse machine operating data, predict possible machine failure, detect unusual behaviour, and support early maintenance decisions. Instead of waiting until a machine completely fails, the system helps users identify possible risks before the condition becomes more serious.

SmartMaint AI uses machine operating features such as air temperature, process temperature, rotational speed, torque, tool wear, and machine type. Before the models are applied, the data is cleaned, prepared, encoded, and scaled. The prepared data is then used by the machine learning models to produce failure predictions and anomaly detection results.

Random Forest is selected as the main supervised machine learning technique in this project. It uses the labelled machine failure information from the AI4I 2020 Predictive Maintenance dataset to learn the relationship between machine operating conditions and failure. Random Forest was chosen as the main model because it achieved the best overall performance during model evaluation.

Isolation Forest is used as a supporting anomaly detection model. It helps identify machine records that are different from normal operating patterns without using the machine failure label directly. This is useful because abnormal and failure records are less common than normal operating records.

One-Class SVM is used as another unsupervised anomaly detection model for comparison. It learns the general pattern of normal machine behaviour and identifies records that fall outside the normal boundary. The comparison between Isolation Forest and One-Class SVM helps show how different unsupervised methods detect unusual machine conditions.

The three models have different roles in SmartMaint AI. Random Forest provides the main machine failure prediction, Isolation Forest provides supporting anomaly evidence, and One-Class SVM is used as an unsupervised comparison model. The model outputs and sensor conditions are then used to support machine health assessment, alert level classification, and maintenance decision-making.

The system does not only display raw model results. It also converts the results into useful maintenance information, including machine health score, alert level, detected problem, probable cause, observed abnormal pattern, recommended action, maintenance schedule suggestion, and maintenance alert log. This makes the system easier to understand because users can quickly identify which machines are healthy, which machines require attention, and what maintenance action should be considered.

2.2.3.1 Random Forest

Random Forest is used in this project as the main supervised machine learning model for machine failure prediction. It combines several decision trees to produce a more stable and reliable prediction. Each decision tree gives its own result, and the final prediction is based on the combined decisions of all the trees. This helps reduce the weakness of depending on only one decision tree.

Unlike Isolation Forest and One-Class SVM, Random Forest requires labelled data during training. This means that the model learns from both the machine operating features and the known machine failure result. In the AI4I 2020 Predictive Maintenance dataset, features such as machine type, air temperature, process temperature, rotational speed, torque, and tool wear are used as the input data. The Machine failure column is used as the target output.

By using this labelled data, Random Forest can learn the relationship between machine operating conditions and possible machine failure. The trained model then produces a failure prediction and failure probability for the testing records. These results are used as the main AI prediction evidence in SmartMaint AI.

Random Forest was selected as the main model because it achieved the best overall performance during the model evaluation. It obtained the highest accuracy, precision, recall, F1-score, ROC-AUC, and PR-AUC compared with Isolation Forest and One-Class SVM. Isolation Forest is still used to provide supporting anomaly evidence, while One-Class SVM is used as an unsupervised comparison model.

Another useful feature of Random Forest is feature importance. Feature importance shows which machine operating features have a greater influence on the model prediction. For example, if torque, rotational speed, or tool wear receives a high importance value, this means that the feature plays an important role in the model's failure predictions.

The feature importance results can help users understand which machine readings should receive more attention. However, feature importance only shows how strongly a feature contributes to the prediction. It does not prove that the feature directly caused the machine failure.

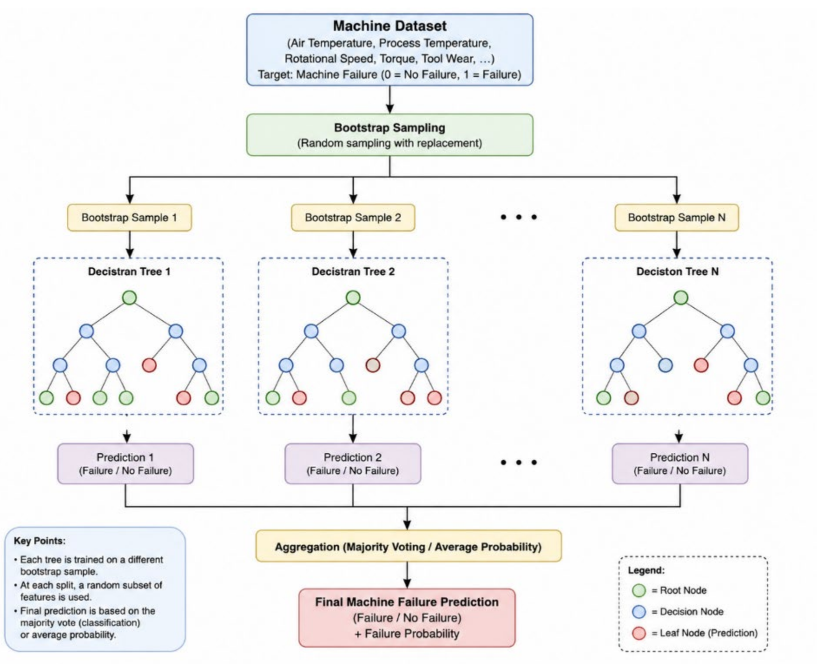


Figure 2.3: Random Forest Model Structure for Machine Failure Prediction

2.2.3.2 Isolation Forest

Isolation Forest is used in this project as a supporting unsupervised anomaly detection model. Its purpose is to identify machine records that are different from the normal operating pattern. In predictive maintenance, unusual machine behaviour may indicate an early sign of equipment degradation or possible failure.

Isolation Forest works by separating unusual records from normal records. Normal machine records are usually more common and follow similar patterns, while abnormal records are less common and are different from the majority of the data. Because of this, Isolation Forest can identify abnormal conditions without depending directly on labelled machine failure data.

This model is useful for predictive maintenance because most machines normally operate without failure, while abnormal and failure conditions occur less often. In some industrial situations, complete failure labels may not be available. Therefore, an unsupervised anomaly detection model can help identify unusual operating behaviour that may require further inspection.

In SmartMaint AI, Isolation Forest analyses machine operating features such as air temperature, process temperature, rotational speed, torque, tool wear, and machine type. The model produces an anomaly label and anomaly score for each record. These results are used as supporting evidence together with the Random Forest prediction, One-Class SVM comparison result, and sensor conditions.

Isolation Forest does not directly determine the final simulated machine health score or alert level. Instead, its results provide additional anomaly evidence that helps the system explain unusual machine behaviour and support maintenance decision-making. Random Forest remains the main machine failure prediction model, while Isolation Forest supports the system by detecting abnormal operating patterns without using the failure label.

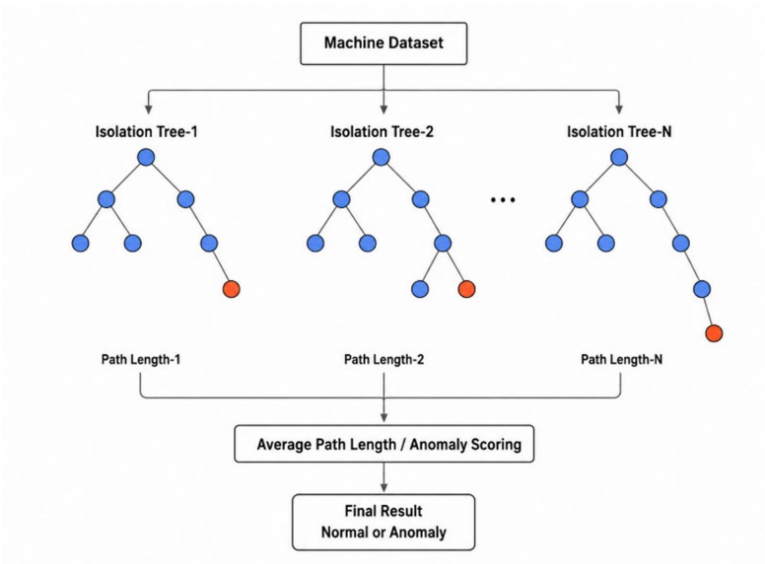


Figure 2.4: Isolation Forest Anomaly Detection Concept

2.2.3.3 One class SVM

One-Class Support Vector Machine, also known as One-Class SVM, is used in this project as an unsupervised comparison model for anomaly detection. This model is useful when the main purpose is to learn the pattern of normal data and identify records that are different from the normal condition.

In predictive maintenance, unusual machine behaviour may appear when sensor readings move outside the normal operating range. One-Class SVM can help identify these unusual records by creating a boundary around the normal machine data. Any record that falls outside this boundary may be classified as abnormal and may indicate an early sign of machine degradation or possible failure.

In SmartMaint AI, One-Class SVM analyses machine operating features such as air temperature, process temperature, rotational speed, torque, tool wear, and machine type. The model learns the general pattern of normal machine operation and produces an anomaly result for each record.

One-Class SVM is not used as the main prediction model in this project. It is used to compare its anomaly detection performance with Isolation Forest. This comparison helps the project understand how different unsupervised models detect unusual machine behaviour.

One limitation of One-Class SVM is that its performance can be affected by parameter settings and data scaling. If the data is not properly standardized or the model parameters are not suitable, the anomaly detection results may become less accurate. For this reason, StandardScaler is applied before using the model.

In the final SmartMaint AI system, Random Forest is used as the main machine failure prediction model, Isolation Forest provides supporting anomaly evidence, and One-Class SVM is kept as an unsupervised comparison model.

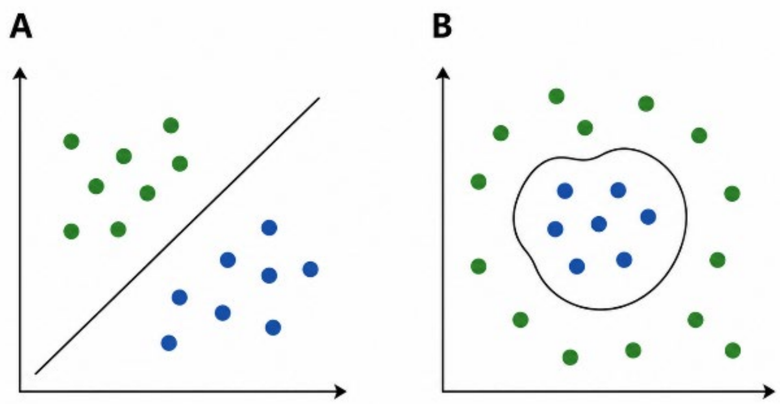


Figure 2.5: One-Class SVM Boundary for Normal and Abnormal Data Detection

2.2.3.4 Machine Health Score and Alert Level

The proposed system does not only display raw machine learning results. The results are also presented as a machine health score and alert level so that users can understand the machine condition more easily. The health score ranges from 0 to 100. A higher score shows a healthier machine condition, while a lower score shows that the machine may require inspection or maintenance.

In SmartMaint AI, the original sensor readings and machine learning outputs are kept as AI prediction evidence. Since the AI4I 2020 Predictive Maintenance dataset does not contain complete machine histories or real fleet information, the system uses controlled simulated machine health conditions for prototype demonstration. This allows the dashboard to show all five maintenance conditions and test the alert, recommendation, scheduling, and reporting functions.

The system classifies the simulated machine condition into five alert levels:

- **Normal:** Health score from 80 to 100
- **Low Risk:** Health score from 60 to below 80
- **Medium Risk:** Health score from 40 to below 60
- **High Risk:** Health score from 20 to below 40
- **Critical:** Health score below 20

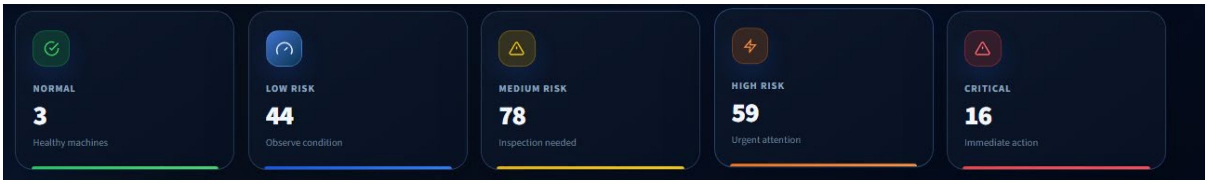


Figure 2.6: The system classifies machines into five alert levels

These alert levels help users quickly identify which machines are operating normally and which machines require observation, inspection, urgent attention, or immediate maintenance action. This makes the system easier to use because users do not need to understand the technical machine learning results directly.
The final simulated health score and alert level are used for dashboard demonstration and maintenance decision support. They should not be considered as direct failure probabilities produced by the Random Forest model.

2.2.3.5 Data Standardisation

Before applying the machine learning models, the sensor data needs to be normalized. This is because the dataset contains different types of values such as temperature, rotational speed, torque, and tool wear. These values have different ranges, so normalization helps make the data more consistent.

$Z = (X - \mu) / \sigma$

Where:

Z = normalized value

X = original sensor value

μ = mean value

σ = standard deviation

2.2.3.6 Risk Evidence and Controlled Health Scenarios

The machine health score is used to show the overall condition of each machine in a simple way. A higher health score means the machine is in a healthier condition, while a lower health score means the machine has a higher maintenance risk and may require inspection or action.

At the record level, the system converts the calculated AI risk into a health score using the following formula:

$$\textbf{Health Score} = \textbf{100} - \textbf{Risk Score}$$

Where:

- **Health Score** = condition score of the analysed record
- **Risk Score** = calculated risk based on Random Forest prediction, supporting anomaly evidence, and abnormal sensor behaviour

In SmartMaint AI, the record-level AI risk is mainly supported by Random Forest, while Isolation Forest provides supporting anomaly evidence and One-Class SVM is used for comparison.

However, the final machine level health score shown in the dashboard is part of the controlled simulated fleet used for prototype demonstration. This allows the system to display all five alert levels consistently and test the dashboard, maintenance alerts, recommendations, scheduling, and report functions.

Therefore, the final simulated machine health score should not be treated as a direct Random Forest failure probability. It is a machine condition score used for system demonstration and maintenance decision support.

2.2.3.7 Maintenance Recommendation

After the machine condition is classified, the SmartMaint AI system generates maintenance information for the user. This information includes the detected problem, probable cause, observed abnormal pattern and recommended maintenance action. This part is to make the machine learning result easier to understand and more useful for maintenance decision making. In many machine learning systems, the output might only show whether the machine is normal or abnormal. However, this is not enough for users who need to decide what action should be taken next. For instance, if the system detects abnormal torque or unusual rotational speed, the user also needs to know what the possible issue is and what inspection should be done. SmartMaint AI therefore makes the prediction result into a more lucid maintenance explanation. The recommendation is generated depending on the machine condition, alert level, equipment type and abnormal sensor pattern.

For instance, if a fan has abnormal rotational speed and high torque, the system might recommend checking the fan blade condition, shaft alignment, bearing condition, or airflow restriction. If a pump has abnormal torque and temperature, it could suggest checking the suction line, impeller, seals, or bearing condition. SmartMaint AI could be useful as a product based system and not only detect machine risk but also make simple decisions on it. The maintenance recommendation tells the user why a machine is considered risky and what should be done with it. It can aid technicians, engineers or maintenance staff in planning inspections before the machine reaches a serious failure point.

2.3 Project Methodology

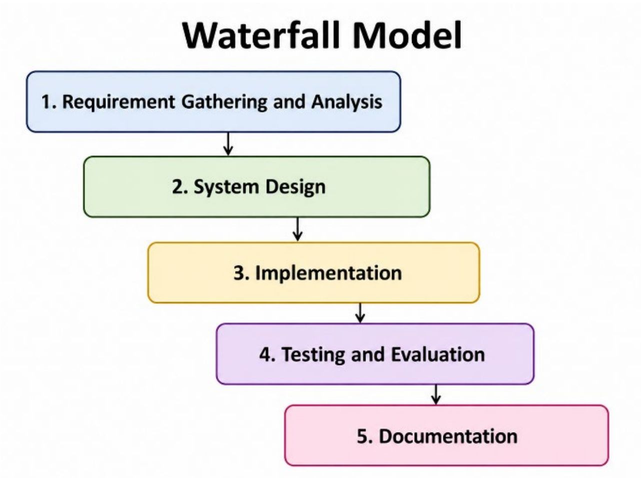


Figure 2.7: Waterfall Model for SmartMaint AI Development

This project uses the waterfall model as the project methodology. The Waterfall Model is chosen because it is able to provide a clear development process from requirement gathering to documentation. Since SmartMaint AI is a product-based project, the system needs to be planned for implementation so that each module can develop and test in the right order. In this project, it is perfect to use the waterfall model because the project requirements are very clear from the beginning. The system needs to load the predictive maintenance data, preprocess the data, apply machine learning models, calculate health scores, classify alert levels, make maintenance recommendations, display the results on a dashboard and give reports. These requirements can be organized in different development phases. The process of this project is structured into five main steps: requirement gathering and analysis, system design, implementation, testing and evaluation, and documentation. In the requirement phase, the project problem, objectives, dataset, and system functions are identified. In the design phase, the system architecture, dashboard layout, and module flow are planned. In the implementation phase, i develop machine learning models and the Streamlit application. The system is then tested and assessed to ensure that the functions work correctly. The project is then documented in the report with explanations, figures, tables, screenshots, and results. This approach helps the project to become more organized and the progress of SmartMaint AI is also tracked, and the final system is to meet the project objectives.

2.3.1 Requirement Gathering and Analysis

Requirement gathering and analysis is the first step in SmartMaint AI development. At this stage, it is necessary to identify a project problem, objectives, scope, target users, dataset and system functions. This phase is important because it helps to know what the system needs to do before the design and implementation process begins. The main problem identified in this project is that machine failures may be detected late, especially when the maintenance process depends only on manual inspection or fixed maintenance timetables. This can cause unexpected downtime, high repair costs and interruption to operation. So, the system is required to detect abnormal machine behavior early in the process using machine learning. The data requirement is also analyses in this stage.

The project uses the AI4I 2020 Predictive Maintenance dataset because it includes machine operation features such as air temperature, process temperature, rotational speed, torque, tool wear and machine failure information. These are required features to train and test machine learning models for predictive maintenance. The functional requirements of the system are also identified. SmartMaint AI should be able to load and process the dataset, run machine learning models, detect abnormal machine behavior, calculate machine health score, classify alert levels, generate maintenance recommendations, display results in a dashboard, show a maintenance alert log, and allow users to export reports. Besides that, software and tools needed for the project are analyses. Python is the main programming language as it supports machine learning and data analysis libraries. Streamlit is used to build the dashboard as it allows for the system to be built as a simple local PC-based application. This phase helps ensure that the project requirements are clear before moving to the system design phase.

2.3.2 System Design

In the system design phase, the overall structure and flow of SmartMaint AI are planned before the implementation process begins. This phase is important because it explains how the system receives machine data, processes the data, applies the machine learning models, and presents the final results to the user. A clear system design also helps ensure that all project modules are properly connected.

The SmartMaint AI process begins with machine operating data from the AI4I 2020 Predictive Maintenance dataset. The dataset contains features such as machine type, air temperature, process temperature, rotational speed, torque, tool wear, and machine failure status. The data is first cleaned and preprocessed before it is used by the machine learning models.

After preprocessing, Random Forest is applied as the main supervised model for machine failure prediction. It produces a failure prediction and failure probability using the labelled machine failure data. Isolation Forest is used as a supporting anomaly detection model to identify unusual operating patterns, while One-Class SVM is used as an unsupervised comparison model.

The original model outputs and sensor readings are kept as AI prediction evidence. Since the AI4I dataset does not contain real fleet identities or complete machine histories, the system also creates a controlled simulated fleet for prototype demonstration. This allows SmartMaint AI to display machine health scores and classify machines into Normal, Low Risk, Medium Risk, High Risk, and Critical conditions.

After the machine condition is determined, the system generates useful maintenance information such as the detected problem, probable cause, observed abnormal pattern, recommended action, maintenance priority, and maintenance schedule suggestion. These results are then displayed through the Streamlit dashboard.

The main modules of SmartMaint AI are the Dashboard Module, Machine Monitoring Module, AI Prediction and Health Scoring Module, Maintenance Alert Log Module, and Report Generation Module. The Dashboard Module gives an overall summary of machine conditions and risk levels. The Machine Monitoring Module allows users to view detailed information for each machine.

The AI Prediction and Health Scoring Module runs the machine learning models, keeps the prediction and anomaly evidence, and prepares the simulated machine level health condition. The Maintenance Alert Log Module displays machines that require observation, inspection, urgent attention, or immediate maintenance. The Report Generation Module allows users to download machine reports and system summaries.

The system design connects the technical machine learning process with practical maintenance information. This allows SmartMaint AI to operate as a complete predictive maintenance application instead of only displaying machine learning predictions.

2.3.3 Implementation

In the implementation phase, the SmartMaint AI system is developed using Python and Streamlit. Python is used as the main programming language because it supports data processing, machine learning, model evaluation, system logic, and report generation. Streamlit is used to build the local PC-based dashboard because it allows the system results to be presented through a simple and interactive interface.

The implementation starts by loading the AI4I 2020 Predictive Maintenance dataset. The data is then cleaned and preprocesses before it is used by the machine learning models. This step is important because the dataset contains different types of machine data, such as machine type, air temperature, process temperature, rotational speed, torque, and tool wear. The categorical values are converted into numerical form, while the numerical features are prepared and standardized where required.

Random Forest is implemented as the main supervised machine learning model for machine failure prediction. The model uses the labelled machine failure data to learn the relationship between machine operating conditions and possible failure. It produces a failure prediction and failure probability for each analysed record.

Isolation Forest is used as a supporting anomaly detection model. It helps identify machine records that are different from the normal operating pattern without using the machine failure label directly. One-Class SVM is also implemented as an unsupervised comparison model to compare another method of detecting unusual machine behaviour.

After the model analysis is completed, the original sensor readings and model outputs are kept as AI prediction evidence. Since the AI4I dataset does not contain complete real machine histories or actual fleet information, the system creates a controlled simulated fleet for prototype demonstration. This allows SmartMaint AI to display machine health scores and classify machines into five conditions: Normal, Low Risk, Medium Risk, High Risk, and Critical.

The system then converts the machine condition into useful maintenance information. This includes the detected problem, probable cause, observed abnormal pattern, recommended action, maintenance priority, maintenance schedule suggestion, and maintenance alert log.

The final results are displayed through the Streamlit dashboard. The dashboard includes the system overview, machine monitoring, maintenance alert log, selected machine details, and report generation functions. Users can view machine conditions, identify machines that require attention, review maintenance recommendations, and download reports.

The implementation phase transforms the machine learning models and decision-support rules into a complete working predictive maintenance application.

2.3.4 Testing and Evaluation

The testing and evaluation process was carried out to check whether SmartMaint AI works correctly and produces useful predictive maintenance results. The evaluation covered both the machine learning models and the main functions of the system.

The machine learning evaluation compared Random Forest, Isolation Forest, and One-Class SVM using suitable performance measures such as accuracy, precision, recall, F1-score, ROC-AUC, and PR-AUC. Random Forest achieved the best overall performance and was selected as the main machine failure prediction model. Isolation Forest was used to provide supporting anomaly evidence, while One-Class SVM was used as an unsupervised comparison model.

The system evaluation also included the machine health score trend, alert level distribution, equipment category summary, high-risk machine distribution, and Random Forest feature importance. The feature importance results show which machine operating features, such as torque, rotational speed, tool wear, air temperature, and process temperature, had a greater influence on the Random Forest prediction.

The health score and alert level results were also tested to make sure that the machine conditions were displayed correctly. The controlled simulated fleet was used to demonstrate all five conditions: Normal, Low Risk, Medium Risk, High Risk, and Critical. This helped test the dashboard, filters, maintenance recommendations, alert log, scheduling, and report generation functions.

The dashboard and report functions were tested by viewing machine details, applying filters, checking maintenance alerts, and downloading reports.

The results showed that SmartMaint AI could present machine learning outputs and simulated machine conditions in a clear and useful way.

testing and evaluation process showed that the system was able to run the machine learning models, display prediction and anomaly evidence, classify simulated machine conditions, generate maintenance information, and support maintenance decision-making.

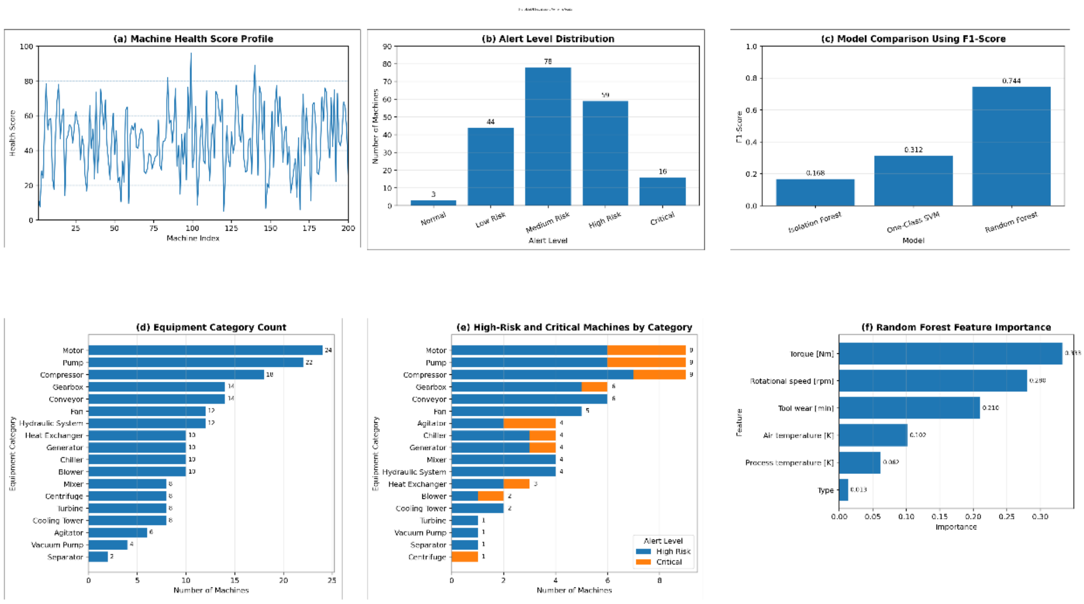


Figure 2.8: SmartMaint AI Evaluation and Monitoring Results

2.4 Project Schedule and Milestones

The project schedule and milestones were planned to ensure that the SmartMaint AI system could be completed within the given Final Year Project period. The schedule organised the project activities in a proper sequence, beginning with project planning and ending with report preparation and final submission.

The main project activities included project title confirmation, problem identification, literature review, dataset study, requirement analysis, system design, machine learning model implementation, dashboard development, system testing, model evaluation, and report documentation. Each activity was arranged according to the project development process.

The first milestone was the confirmation of the project title, objectives, and scope. The literature review was then conducted to study predictive maintenance, machine learning techniques, rotating equipment, and related existing systems.

The AI4I 2020 Predictive Maintenance dataset was studied and prepared before model development. This stage included understanding the dataset features, cleaning the data, encoding machine type values, removing unnecessary columns, and standardising numerical features where required.

After data preparation, the machine learning models were implemented and evaluated. Random Forest was developed as the main supervised machine failure prediction model. Isolation Forest was implemented as a supporting unsupervised anomaly detection model, while One-Class SVM was used as an unsupervised comparison model.

The controlled simulated fleet and rule-based maintenance functions were then developed to demonstrate health scores, alert levels, maintenance priorities, detected problems, recommended actions, alert logs, and maintenance schedule suggestions.

The SmartMaint AI dashboard was developed using Streamlit to display the machine learning results, simulated machine conditions, maintenance information, and downloadable reports. The final milestones included functional testing, model evaluation, dashboard testing, report writing, correction of the project documentation, and preparation for the final project presentation.

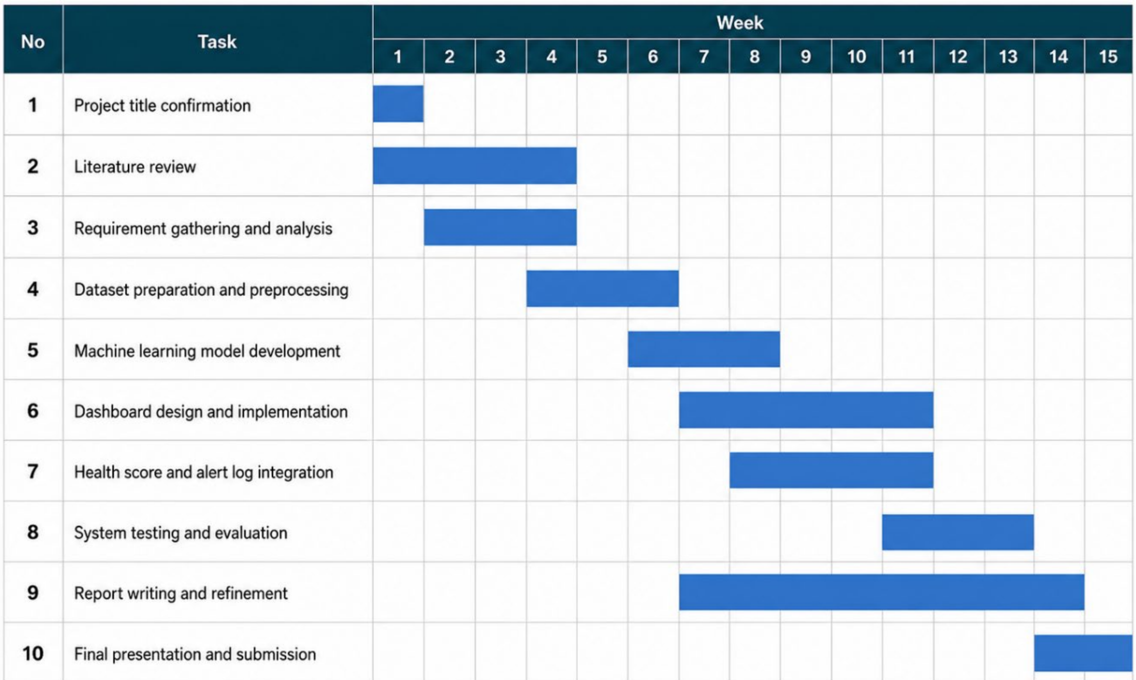


Figure 2.9: Project Schedule and Milestones for SmartMaint AI

2.5 Conclusion

This chapter discussed the literature review and project methodology used for SmartMaint AI. It explained the importance of predictive maintenance and rotating equipment monitoring in helping users identify possible machine problems before serious failure occurs. The chapter also discussed how machine learning can be used to analyse machine operating data and detect unusual behaviour.

The existing maintenance approaches, including reactive maintenance and preventive maintenance, were reviewed together with their limitations. Reactive maintenance only takes place after failure occurs, while preventive maintenance follows a fixed schedule and may cause unnecessary maintenance. Predictive maintenance provides a better approach by using machine data to support maintenance decisions.

The proposed machine learning techniques were also explained. Random Forest is used as the main supervised model for machine failure prediction because it achieved the best overall evaluation results. Isolation Forest is used as a supporting unsupervised anomaly detection model, while One-Class SVM is used as an unsupervised comparison model.

This chapter also explained the project methodology based on the Waterfall Model. The main phases include requirement gathering and analysis, system design, implementation, testing and evaluation, and documentation. Each phase provides a clear development process for building the SmartMaint AI system.

Overall, this chapter provided the background, related concepts, proposed techniques, and development approach used in the project. The next chapter discusses the system analysis, including problem analysis, data requirements, functional requirements, non-functional requirements, and other system requirements.

CHAPTER 3: REQUIREMENT ANALYSIS

3.1 Introduction

This chapter discusses the requirement analysis of the SmartMaint AI system. Requirement analysis is important because it identifies what the system needs before the design and implementation phases begin. A clear requirement analysis helps ensure that all system functions, data, software, and hardware requirements are properly planned.

This chapter covers the problem analysis, data requirements, functional requirements, non-functional requirements, software requirements, and hardware requirements of the project. SmartMaint AI is a local PC-based predictive maintenance application developed for rotating equipment.

The system uses machine learning to analyse machine operating data and support maintenance decision-making. Random Forest is used as the main supervised model for machine failure prediction. Isolation Forest is used to provide supporting anomaly detection evidence, while One-Class SVM is used as an unsupervised comparison model.

The system also presents machine conditions using health scores and alert levels. It generates maintenance information such as detected problems, probable causes, observed abnormal patterns, recommended actions, maintenance schedules, alert logs, and machine reports.

Since the AI4I 2020 Predictive Maintenance dataset does not contain complete real machine histories or fleet information, SmartMaint AI also uses a controlled simulated fleet for prototype demonstration. This allows the system to test and display the dashboard, machine monitoring, alert levels, maintenance recommendations, scheduling, and report generation functions.

This chapter also describes the resources required to develop and run the system, including the AI4I 2020 Predictive Maintenance dataset, Python libraries, the Streamlit dashboard, and computer hardware. Overall, the requirement analysis provides a clear foundation for the system design and implementation phases.

3.2 Problem Analysis

Rotating equipment such as motors, pumps, fans, compressors, generators, and gearboxes plays an important role in many industrial operations. These machines are commonly used in manufacturing, production, and other industrial processes. When one of these machines fails unexpectedly, it may interrupt the whole operation, increase repair costs, delay production, and create safety risks. Therefore, monitoring machine condition is important so that possible problems can be identified before they become serious.

The first problem is that many maintenance activities still depend on reactive maintenance. In reactive maintenance, the machine is repaired only after it fails or stops operating. Although this method is simple, it is not effective because the machine may already be seriously damaged when the problem is discovered. This may result in higher repair costs, longer downtime, and urgent maintenance work.

The second problem is the limitation of preventive maintenance. Preventive maintenance is carried out according to a fixed schedule, such as weekly, monthly, or after a certain number of operating hours. This approach is better than waiting for a machine to fail, but it does not always represent the actual condition of the machine. Some machines may still be healthy but are serviced too early, which causes unnecessary maintenance work and costs. Other machines may develop faults before the scheduled maintenance date.

Manual machine inspection also depends heavily on the knowledge and experience of technicians. Different technicians may understand the same machine condition differently. This may lead to different decisions, late detection, or missed warning signs. When many machines must be monitored, it is also difficult for maintenance staff to inspect every machine continuously.

Machine operating data can also be difficult to understand without proper analysis. Sensor values such as air temperature, process temperature, rotational speed, torque, and tool wear may change during machine operation. Some changes may be normal, while others may indicate abnormal behaviour or possible machine failure. Small abnormal patterns may not be easy to identify through manual checking.

A basic prediction result alone may also be difficult for users to understand. Maintenance users need clear and practical information, such as the machine health score, alert level, detected problem, probable cause, observed abnormal pattern, recommended action, and maintenance schedule suggestion. Without this information, the machine learning result may not be useful for maintenance decision-making.

To address these problems, SmartMaint AI is proposed as a local PC-based predictive maintenance application. The system uses machine learning to analyse machine operating data and provide useful maintenance information. Random Forest is used as the main supervised model for machine failure prediction because it achieved the best overall evaluation results. Isolation Forest is used to provide supporting anomaly detection evidence, while One-Class SVM is used as an unsupervised comparison model.

The model results and machine operating conditions are then presented through health scores, alert levels, detected problems, probable causes, observed abnormal patterns, recommended actions, maintenance schedule suggestions, alert logs, and machine reports. A controlled simulated fleet is also used to demonstrate the machine-level dashboard and all five alert conditions.

SmartMaint AI helps users identify which machines are operating normally and which machines require attention. The Streamlit dashboard presents the results in a simple form so that students, technicians, engineers, and maintenance personnel can understand machine conditions and make earlier maintenance decisions.

3.3 Requirement Analysis

Requirement analysis is carried out to identify the main needs of the SmartMaint AI system before the design and implementation phases begin. This analysis helps ensure that the system can achieve the project objectives and perform all required functions correctly.

Since SmartMaint AI is a predictive maintenance application, its requirements must cover both the machine learning process and the user interface. The main requirements include data requirements, functional requirements, non-functional requirements, software requirements, and hardware requirements.

SmartMaint AI requires machine operating data to analyse machine conditions and support early maintenance decisions. The system uses the AI4I 2020 Predictive Maintenance dataset, which contains important features such as machine type, air temperature, process temperature, rotational speed, torque, tool wear, and machine failure status.

The machine failure status is used as the target output for training and evaluating the Random Forest model. Random Forest is the main supervised model used for machine failure prediction. Isolation Forest is used to provide supporting anomaly detection evidence, while One-Class SVM is used as an unsupervised comparison model.

The system also requires controlled simulated machine-level data because the AI4I dataset does not provide complete real machine histories or an actual equipment fleet. The simulated fleet allows the system to demonstrate machine health scores, alert levels, maintenance recommendations, scheduling, alert logs, and report generation.

The functional requirements describe the tasks that the system must perform. SmartMaint AI should be able to load the dataset, clean and preprocess the data, train and run the machine learning models, evaluate model performance, and display prediction and anomaly results.

The system should also generate machine health scores, classify machines into Normal, Low Risk, Medium Risk, High Risk, and Critical conditions, display machine monitoring information, provide maintenance recommendations, generate maintenance schedule suggestions, show the maintenance alert log, and allow users to download reports.

The non-functional requirements describe the expected quality of the system. SmartMaint AI should be simple, reliable, understandable, and easy to use. The system is designed for users such as students, technicians, engineers, and maintenance personnel who may not have advanced knowledge of machine learning.

Therefore, the Streamlit dashboard should present the results clearly through health scores, alert levels, charts, tables, maintenance recommendations, and machine reports. This helps users quickly identify which machines are operating normally and which machines require attention.

This section explains the requirements needed to develop SmartMaint AI as a local PC-based predictive maintenance application. The following subsections describe the data requirements, functional requirements, non-functional requirements, software requirements, and hardware requirements in more detail.

3.3.1 Data Requirement

Data requirements are important for SmartMaint AI because the system depends on machine operating data to perform predictive maintenance analysis. The quality and suitability of the data can affect the machine learning process and the reliability of the results produced by the system.

This project uses the AI4I 2020 Predictive Maintenance dataset for training, testing, and evaluating the machine learning models. The dataset contains 10,000 records and includes important machine features such as machine type, air temperature, process temperature, rotational speed, torque, and tool wear. It also contains the Machine failure column, which shows whether a failure occurred.

These features describe different parts of the machine operating condition. For example, temperature values provide information about the thermal condition, rotational speed represents machine speed, torque represents the operating load, and tool wear shows the amount of usage or wear. Changes in these values may provide useful information for detecting unusual behaviour or predicting machine failure. However, a single abnormal reading does not always prove that a failure has occurred.

The data is used in several stages of SmartMaint AI. First, the dataset is loaded and checked for missing values, incorrect data types, and unnecessary columns. During preprocessing, the machine type is converted into numerical form, while the numerical features are standardized where required by the model.

Random Forest uses the machine operating features as input and the Machine failure column as the labelled target. It is trained as the main supervised model for machine failure prediction. Isolation Forest analyses the operating features without using the failure label and provides supporting anomaly evidence. One-Class SVM also uses the prepared operating features as an unsupervised comparison model.

The system produces record-level outputs such as the Random Forest failure prediction, failure probability, Isolation Forest anomaly result, anomaly score, and One-Class SVM comparison result. These outputs are kept as machine learning evidence and are used together with the original sensor readings to explain the analysed machine condition.

The AI4I dataset does not provide complete machine histories, real equipment identities, or an actual industrial fleet. Therefore, SmartMaint AI also requires controlled simulated machine-level data for prototype demonstration. The simulated fleet contains 200 machines from 18 equipment categories. Each machine is given a demonstrated health score and one of five alert levels: Normal, Low Risk, Medium Risk, High Risk, or Critical.

The system also generates maintenance-related output data, including the detected problem, probable cause, observed abnormal pattern, recommended action, maintenance priority, maintenance schedule suggestion, and alert log information. These outputs convert the technical analysis into information that users can understand more easily.

Overall, the main data required by SmartMaint AI includes the original machine operating features, machine failure labels, model prediction outputs, anomaly detection outputs, controlled simulated machine conditions, health scores, alert levels, and maintenance recommendation data. These data requirements allow the system to present useful information through the dashboard, maintenance alert log, and generated reports.

Table 3.1: Data Requirement

Data	Description	Purpose
Machine type	Product quality category represented as L, M, or H	Used as an input feature for machine learning analysis
Air temperature	Temperature of the surrounding air in Kelvin	Used to analyse the thermal operating condition of the machine
Process temperature	Temperature recorded during the machine process in Kelvin	Used to identify unusual process heat conditions

Rotational speed	Speed of machine rotation measured in revolutions per minute	Used to analyse changes or instability in machine speed
Torque	Rotational force or operating load applied to the machine	Used to analyse machine load and possible mechanical stress
Tool wear	Total tool usage or wear time measured in minutes	Used to represent machine usage and possible degradation
Machine failure	Label showing whether machine failure occurred	Used as the target output for training and evaluating the Random Forest model
Random Forest prediction	Predicted machine failure class and failure probability	Used as the main machine failure prediction evidence
Isolation Forest output	Anomaly label and anomaly score produced without using the failure target	Used as supporting evidence for unusual machine behaviour
One-Class SVM output	Normal or abnormal result produced by the comparison model	Used to compare unsupervised anomaly detection performance
Equipment category	Simulated rotating-equipment type, such as motor, pump, fan, or compressor	Used to organise the 200-machine simulated fleet
Health score	Machine condition score ranging from 0 to 100	Used to present the simulated machine condition in a simple form
Alert level	Machine condition classified as Normal, Low Risk, Medium Risk, High Risk, or Critical	Used to show the level of maintenance attention required
Detected problem	Maintenance issue identified from the machine condition and sensor pattern	Used to explain the possible machine problem

Probable cause	Possible reason related to the detected problem	Used to support maintenance inspection and decision-making
Observed abnormal pattern	Summary of unusual sensor or model evidence	Used to explain why the machine may require attention
Recommended action	Suggested maintenance or inspection activity	Used to guide the user in responding to the machine condition
Maintenance schedule suggestion	Suggested maintenance type, urgency, status, due date, and reason	Used to support maintenance planning

3.3.2 Functional Requirement

Functional requirements describe the main functions that the SmartMaint AI system must perform. These requirements explain what the system should do to support predictive maintenance and help users understand machine conditions. Since this project is a product-based system, the functional requirements cover both the machine learning process and the dashboard functions.

The system should be able to load the AI4I 2020 Predictive Maintenance dataset and check that the required data is available. It should clean and preprocess the data by removing unnecessary columns, converting machine type values into numerical form, and standardizing the numerical features where required.

SmartMaint AI should run Random Forest as the main supervised machine learning model for machine failure prediction. The system should generate a predicted failure class and failure probability for the analysed records. Isolation Forest should also be used to provide supporting anomaly labels and anomaly scores, while One-Class SVM should be used as an unsupervised comparison model.

The system should evaluate the performance of the three machine learning models using suitable measures such as accuracy, precision, recall, F1-score, ROC-AUC, and PR-AUC. These evaluation results should help explain why Random Forest is selected as the main prediction model.

The system should also generate a controlled simulated fleet because the AI4I dataset does not contain complete machine histories or a real equipment fleet. The simulated fleet should contain 200 machines from 18 equipment categories and should support the demonstration of all five alert conditions: Normal, Low Risk, Medium Risk, High Risk, and Critical.

After the machine analysis is completed, SmartMaint AI should present the machine condition through a health score and alert level. It should also generate useful maintenance information, including the detected problem, probable cause, observed abnormal pattern, recommended action, maintenance priority, and maintenance schedule suggestion.

The dashboard should display an overall summary of machine conditions, alert-level distribution, equipment categories, health-score information, and active maintenance alerts. Users should also be able to select an individual machine and view its detailed operating information, machine condition, maintenance recommendation, and schedule suggestion.

The Maintenance Alert Log should display machines that require attention and allow users to review them according to their alert level, equipment category, priority, or maintenance condition. The system should also support report generation so that users can download a clear maintenance summary for the selected machine.

These functional requirements ensure that SmartMaint AI operates as a complete predictive maintenance application rather than only producing machine learning predictions. The system combines data processing, machine learning, machine monitoring, maintenance decision support, dashboard visualisation, alert logging, scheduling, and report generation in one local PC-based application.

Table 3.2: Functional Requirement

No.	Functional Requirement	Description
FR1	Load Dataset	The system should load the AI4I 2020 Predictive Maintenance dataset for analysis.
FR2	Data Preprocessing	The system should clean the dataset, remove unnecessary columns, convert machine type values into numerical form, and standardize numerical features where required.
FR3	Run AI Analysis	The system should run the machine learning analysis locally using Python.
FR4	Machine Failure Prediction	The system should use Random Forest as the main supervised model to predict machine failure and produce failure probability.
FR5	Supporting Anomaly Detection	The system should use Isolation Forest to produce anomaly labels and anomaly scores as supporting evidence of unusual machine behaviour.
FR6	Unsupervised Model Comparison	The system should use One-Class SVM as an unsupervised comparison model for anomaly detection.
FR7	Model Evaluation	The system should compare the models using accuracy, precision, recall, F1-score, ROC-AUC, and PR-AUC.

FR8	Simulated Fleet Generation	The system should generate a controlled simulated fleet of 200 machines from 18 equipment categories for prototype demonstration.
FR9	Health Score Preparation	The system should provide a machine health score ranging from 0 to 100 for each simulated machine condition.
FR10	Alert Level Classification	The system should classify each simulated machine as Normal, Low Risk, Medium Risk, High Risk, or Critical.
FR11	Machine Monitoring	The system should display machine ID, equipment category, health score, alert level, operating data, and maintenance information.
FR12	Maintenance Recommendation	The system should provide the detected problem, probable cause, observed abnormal pattern, and recommended action.
FR13	Maintenance Schedule Suggestion	The system should provide maintenance type, urgency, schedule status, suggested due date, and reason.
FR14	Maintenance Alert Log	The system should display machines that require attention and allow users to review them by alert level, equipment category, priority, and maintenance condition.
FR15	Dashboard Visualisation	The system should display KPI cards, charts, tables, health summaries, alert distributions, and equipment-category information.
FR16	Machine Report Generation	The system should allow users to download a maintenance report for a selected machine.
FR17	System Summary Report	The system should allow users to download a summary report containing overall machine and alert information.

3.3.3 Non Functional Requirement

Non-functional requirements describe how the SmartMaint AI system should perform rather than the specific functions it provides. These requirements focus on the quality of the system, including usability, performance, reliability, maintainability, portability, readability, availability, and data security.

SmartMaint AI should be easy to use because some target users may not have advanced knowledge of machine learning. The Streamlit dashboard should present machine conditions clearly using KPI cards, charts, tables, health scores, alert levels, maintenance recommendations, and reports. Users should be able to understand which machines are operating normally and which machines require attention without interpreting the technical model outputs directly.

The system should also process the dataset and display the results within a reasonable time on a personal computer. Since SmartMaint AI is a local PC-based application, it should not require a cloud server or a continuous internet connection after the required software, dataset, and project files have been installed.

The system should produce consistent results when the same dataset, preprocessing steps, model parameters, and simulation settings are used. The source code should also be organised into separate modules so that errors can be corrected and future improvements can be added more easily.

These non-functional requirements help ensure that SmartMaint AI is not only able to perform its required functions but is also practical, understandable, reliable, and suitable for project demonstration and future development.

Table 3.3: Non Functional Requirement

No.	Non-Functional Requirement	Description
NFR1	Usability	The system should provide a simple and clear interface so that users can understand machine conditions and maintenance information easily.
NFR2	Performance	The system should process the dataset, run the machine learning analysis, and display the results within a reasonable time on a personal computer.
NFR3	Reliability	The system should produce consistent outputs when the same dataset, preprocessing steps, model settings, and simulation settings are used.

NFR4	Maintainability	The source code should be organised into separate modules so that errors can be corrected and future improvements can be added easily.
NFR5	Portability	The system should be able to run locally on a compatible Windows computer with Python and the required libraries installed.
NFR6	Readability	The system outputs should be clearly presented, including the health score, alert level, detected problem, probable cause, observed abnormal pattern, recommended action, and maintenance schedule suggestion.
NFR7	Availability	The system should remain accessible locally as long as the required dataset, project files, Python environment, and libraries are available.
NFR8	Data Security	The system should process and store its data locally and should not expose sensitive information, passwords, or API keys.
NFR9	Compatibility	The dashboard should work correctly through a supported web browser when the Streamlit application is running on the local computer.

3.3.4 Software Requirement

Software requirements describe the software, tools, and libraries needed to develop and run the SmartMaint AI system. Since this project uses machine learning and a dashboard interface, several Python libraries are required for data processing, model training, visualization, and application development.
python is used as the main programming language because it supports many machine learning and data analysis libraries. Streamlit is used to develop the local PC-based dashboard because it is simple, interactive, and suitable for presenting machine learning results.

Table 3.4: Software Requirement

Software / Library	Usage
Windows Operating System	Used as the main operating system for developing and running the project.
Python	Main programming language used for data processing, machine learning, and system development.
Streamlit	Used to build the dashboard and user interface for SmartMaint AI.
Pandas	Used to load, clean, process, and manage dataset files.
NumPy	Used for numerical calculations and data handling.
Scikit learn	Used to implement machine learning models such as Isolation Forest, One class SVM, and Random Forest.
Plotly	Used to create interactive charts and visualizations in the dashboard.
VS Code	Used as the code editor for writing and managing project files.
Microsoft Word	Used to prepare the Final Year Project report.

3.3.5 Hardware Requirement

Hardware requirements describe the physical devices needed to develop and run the SmartMaint AI system. Because this project is a local PC-based application, the main hardware required is a personal computer or laptop that can run python, Streamlit, and the required machine learning libraries. The system does not require IoT sensors or real-time industrial hardware at this stage because the project uses the AI4I 2020 Predictive Maintenance dataset. Therefore, all analysis, model training, dashboard display, and report generation can be done on a regular computer.

3.4 Conclusion

This chapter discussed the requirement analysis of the SmartMaint AI system. It explained the problem analysis, data requirements, functional requirements, non-functional requirements, software requirements, and hardware requirements needed to develop and operate the system.

SmartMaint AI requires machine operating data from the AI4I 2020 Predictive Maintenance dataset. Random Forest is used as the main supervised model for machine failure prediction, while Isolation Forest provides supporting anomaly evidence and One-Class SVM is used as an unsupervised comparison model.

The chapter also described the main system functions, including data preprocessing, model evaluation, simulated fleet generation, health score preparation, alert-level classification, machine monitoring, maintenance recommendations, maintenance schedule suggestions, alert logging, dashboard visualisation, and report generation.

The non-functional requirements ensure that the system is usable, reliable, maintainable, portable, readable, secure, and able to perform within a reasonable time. The required software and hardware resources were also identified to support the development and local operation of the system.

CHAPTER 4: DESIGN

4.1 Introduction

This chapter describes the design of the SmartMaint AI system. The design phase is important because it explains how the system components are organised and how data moves through the system before the final results are presented to the user. SmartMaint AI is a local PC-based predictive maintenance application that uses machine learning to analyse machine operating data and support early maintenance decisions.

The system design begins with data from the AI4I 2020 Predictive Maintenance dataset. The main processes include data loading, data preprocessing, machine learning analysis, model evaluation, simulated fleet generation, machine health score preparation, alert-level classification, maintenance recommendation, maintenance scheduling, alert logging, dashboard visualisation, and report generation.

Random Forest is designed as the main supervised machine learning model for machine failure prediction. It produces a predicted failure class and failure probability using the labelled machine failure data. Isolation Forest is used as a supporting unsupervised model to identify unusual machine behaviour, while One-Class SVM is used as an unsupervised comparison model.

The system does not only display raw model outputs. The prediction results, anomaly evidence, sensor readings, and controlled simulated machine conditions are presented as useful maintenance information. This information includes the machine health score, alert level, detected problem, probable cause, observed abnormal pattern, recommended action, maintenance priority, maintenance schedule suggestion, alert log, and report summary.

Since the AI4I dataset does not provide a complete real equipment fleet or full machine histories, the system design also includes a controlled simulated fleet of 200 machines from 18 equipment categories. This simulated fleet is used to demonstrate machine monitoring, all five alert levels, maintenance recommendations, scheduling, alert logs, and report generation.

The system design is divided into several parts, including system architecture, system flow, module design, user interface design, and output file design. These design elements help ensure that SmartMaint AI operates as a complete predictive maintenance application rather than only as a machine learning experiment.

4.2 System Architecture

The architecture of the SmartMaint AI system shows how the project components are organised and connected. It explains how machine data moves from the input stage through data processing and machine learning analysis before the final maintenance information is presented to the user.

The system begins with the AI4I 2020 Predictive Maintenance dataset. The dataset contains machine operating features such as machine type, air temperature, process temperature, rotational speed, torque, and tool wear. It also includes the Machine failure column, which is used as the target output for training and evaluating the Random Forest model.

The next component is data preprocessing. In this stage, the dataset is checked and prepared before being used by the machine learning models. The preprocessing process includes removing unnecessary columns, checking missing values and data types, converting categorical machine type values into numerical form, selecting the required features, and standardizing numerical values where required. Proper preprocessing is important because the quality of the input data can affect the machine learning results.

After preprocessing, the prepared data is passed to the machine learning analysis component. Random Forest is used as the main supervised model for machine failure prediction. It generates a predicted failure class and failure probability based on the labelled machine failure data.

Isolation Forest is used as a supporting unsupervised anomaly detection model. It produces an anomaly label and anomaly score to provide additional evidence of unusual machine behaviour. One-Class SVM is also used as an unsupervised comparison model. The performance of the three models is evaluated using accuracy, precision, recall, F1-score, ROC-AUC, and PR-AUC.

The system then keeps the original sensor readings and model outputs as AI prediction evidence. Since the AI4I dataset does not contain complete machine histories or a real industrial fleet, the architecture also includes a controlled simulated fleet of 200 machines from 18 equipment categories. This fleet is used to demonstrate machine level health scores, alert levels, maintenance recommendations, scheduling, alert logs, and reports.

The result interpretation and decision-support component converts the machine condition into useful maintenance information. This includes the machine health score, alert level, detected problem, probable cause, observed abnormal pattern, recommended action, maintenance priority, and maintenance schedule suggestion.

Finally, the results are displayed through the Streamlit user interface. Users can view the system overview, monitor individual machine conditions, check alert levels, review maintenance alert logs, view maintenance recommendations, and download machine or system summary reports.

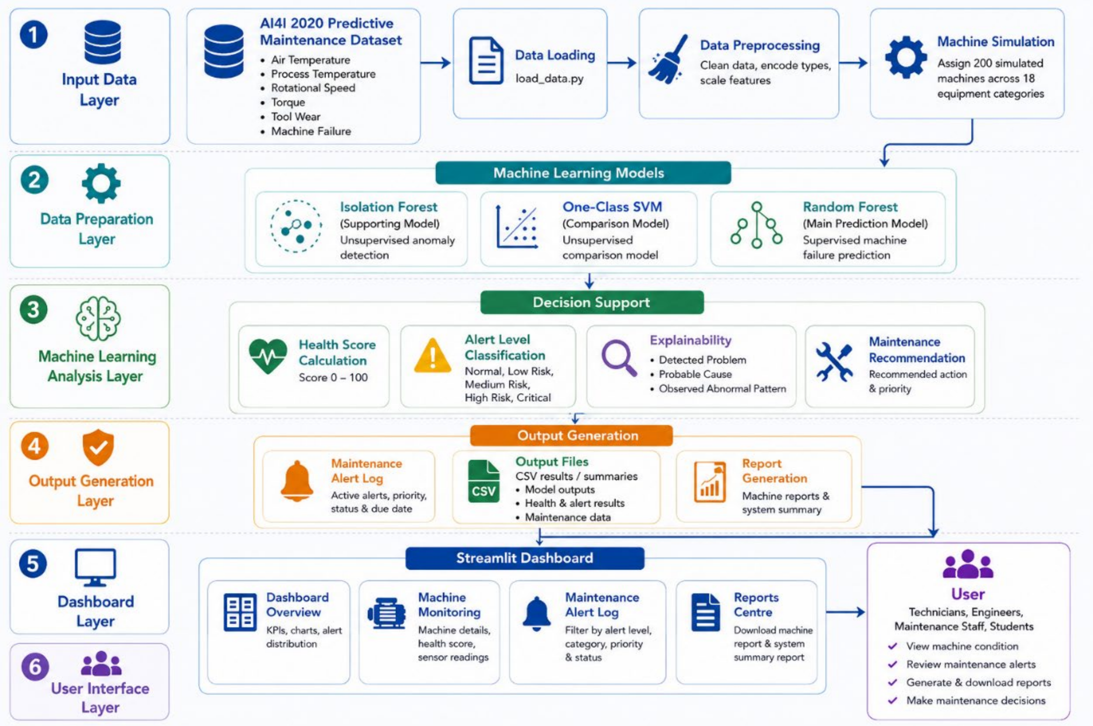


Figure 4.1: Overall System Architecture of SmartMaint AI

4.3 System Flow

The system flow of SmartMaint AI explains how machine operating data is processed and converted into useful maintenance information. The process begins when the AI4I 2020 Predictive Maintenance dataset is loaded into the system. The dataset contains machine operating features such as machine type, air temperature, process temperature, rotational speed, torque, tool wear, and machine failure status.

After the dataset is loaded, the data preprocessing stage is performed. During this stage, unnecessary columns are removed, the required features are selected, categorical machine type values are converted into numerical form, and numerical features are standardized where required. This step ensures that the data is properly prepared before it is passed to the machine learning models.

The prepared records are then assigned to a controlled simulated fleet of 200 machines from 18 equipment categories. Each simulated machine contains 50 machine operating records. This process allows the system to demonstrate machine-level monitoring because the original AI4I dataset does not provide complete machine identities, equipment categories, or operating histories.

The next stage is the machine learning analysis. Random Forest is used as the main supervised model for machine failure prediction. It analyses the prepared machine features and produces a predicted failure class and failure probability. Isolation Forest is used as a supporting unsupervised model to produce anomaly labels and anomaly scores. One-Class SVM is used as an unsupervised comparison model.

The models are evaluated using accuracy, precision, recall, F1-score, ROC-AUC, and PR-AUC. Random Forest achieved the best overall performance and is therefore used as the main prediction model. The Isolation Forest result remains supporting anomaly evidence, while One-Class SVM provides an additional comparison of abnormal-pattern detection.

After the model analysis, the original sensor readings and model outputs are kept as AI evidence. The system then prepares the controlled machine-level health conditions used for prototype demonstration. Each simulated machine receives a health score from 0 to 100 and is classified as Normal, Low Risk, Medium Risk, High Risk, or Critical.

The decision-support stage generates practical maintenance information based on the machine condition and available evidence. This information includes the detected problem, probable cause, observed abnormal pattern, recommended action, maintenance priority, and maintenance schedule suggestion.

Finally, the results are displayed through the Streamlit dashboard. Users can view the overall fleet condition, monitor individual machines, review sensor readings, check the maintenance alert log, examine maintenance recommendations, and download machine or system summary reports.

Overall, the system flow allows SmartMaint AI to convert machine operating data and machine learning evidence into clear maintenance decision-support information. The final simulated health score and alert level are used for prototype demonstration and should not be considered direct Random Forest failure probabilities.

Figure 4.2: System Flow of SmartMaint AI

4.4 Module Design

The SmartMaint AI system is divided into several modules. Each module performs a specific function and supports the predictive maintenance process. Dividing the system into modules makes it easier to develop, understand, test, maintain, and improve in the future.

The first module is the **Dashboard Module**. This module provides an overall summary of the simulated machine fleet. It displays information such as the total number of machines, average health score, alert-level distribution, equipment-category information, and machines that require attention. It also presents KPI cards, charts, and summary tables. This allows users to understand the overall machine condition quickly.

The second module is the **Machine Monitoring Module**. This module allows users to select and view the details of an individual machine. The information includes the machine ID, equipment category, health score, alert level, sensor readings, detected problem, probable cause, observed abnormal pattern, recommended action, maintenance priority, and maintenance schedule suggestion. This module helps users identify the condition of a specific machine and understand why it may require attention.

The third module is the **AI Prediction and Health Scoring Module**. This module manages the machine learning and machine-condition analysis processes. It loads and preprocesses the AI4I 2020 Predictive Maintenance dataset before running the machine learning models.

Random Forest is used as the main supervised model for machine failure prediction. It produces a predicted failure class and failure probability. Isolation Forest is used as a supporting unsupervised anomaly detection model, while One-Class SVM is used as an unsupervised comparison model. The module also evaluates the models using suitable performance measures.

This module also supports the controlled simulation of 200 machines from 18 equipment categories. It prepares the machine-level health scores and classifies each simulated machine as Normal, Low Risk, Medium Risk, High Risk, or Critical. The simulated condition is used to demonstrate the monitoring, maintenance, alert, scheduling, and reporting functions.

The fourth module is the **Maintenance Alert Log Module**. This module displays machines that require observation, inspection, urgent attention, or immediate maintenance. It shows information such as machine ID, equipment category, health score, alert level, priority, maintenance schedule status, and suggested due date. Users can review the alerts using filters such as alert level, equipment category, priority, or maintenance condition.

The fifth module is the **Report Generation Module**. This module allows users to download maintenance information for documentation and future reference. The selected machine report includes machine information, health summary, detected problem, observed abnormal pattern, recommended action, maintenance schedule suggestion, and a short AI summary. The system can also generate a summary report containing overall fleet and alert information.

Overall, these modules work together to convert machine operating data and machine learning evidence into understandable maintenance information. The modular design makes SmartMaint AI more suitable as a complete local PC-based predictive maintenance application.

Table 4.1: SmartMaint AI System Modules and Descriptions

Module	Description
Dashboard Module	Displays overall machine health, alert distribution, KPI cards, and charts.
Machine Monitoring Module	Displays detailed machine information and maintenance condition.
AI Prediction and Health Scoring Module	Runs machine learning models, detects anomalies, calculates health score, and classifies alert level.
Maintenance Alert Log Module	Shows machines requiring attention, inspection, or urgent maintenance.
Report Generation Module	Generates downloadable maintenance summaries and reports.

4.5 User Interface Design

The user interface of SmartMaint AI is built with Streamlit. Its goal is to present machine learning results in a straightforward way. The target users include technicians, engineers, maintenance staff, and students. Therefore, the interface must show machine condition clearly without requiring a deep understanding of technical machine learning outputs.

The user interface consists of several main pages: the Dashboard page, Machine Monitoring page, Maintenance Alert Log page, and Reports page. Each page serves a different purpose, but they all connect to the same predictive maintenance output. The interface uses KPI cards, tables, charts, alert levels, and downloadable reports to help users grasp the condition of the machines.

The Dashboard page gives a summary of the system. It displays key information such as the total number of machines, alert level distribution, health condition summary, and high risk machines. This page helps users quickly recognize the overall condition of all machines.

The Machine Monitoring page lets users see detailed information for each machine. It shows machine ID, equipment category, health score, alert level, detected problem, probable cause, observed abnormal pattern, and recommended maintenance action. This page is essential because it helps users focus on individual machines that may need inspection or maintenance.

The Maintenance Alert Log page lists active alerts generated by the system. It clearly highlights machines with Medium Risk, High Risk, or Critical conditions so users can determine which machines need immediate attention. This page aids in maintenance planning by helping users prioritize machines. The Reports page enables users to create and export maintenance reports. Each report includes machine information, health summary, detected problem, observed abnormal pattern, recommended action, and a brief AI summary. This function is useful for documentation, presentation, and maintenance record keeping.

4.5.1 Dashboard interface

Figure 4.3 shows the Dashboard interface of the SmartMaint AI system. This page provides an overall view of the controlled simulated machine fleet, including machine health conditions, alert-level distribution, equipment categories, and machines that require maintenance attention.

At the top of the dashboard, the system displays general information such as the AI analysis pipeline, system status, and last updated time. Random Forest is identified as the main supervised model for machine failure prediction. Isolation Forest provides supporting anomaly evidence, while One-Class SVM is used as an unsupervised comparison model.

The dashboard also contains several key performance indicator cards. These cards show the total number of simulated machines and the number of machines classified under each alert level: Normal, Low Risk, Medium Risk, High Risk, and Critical. The final simulated fleet contains 200 machines from 18 equipment categories.

The visualisation section presents charts showing the overall machine condition, alert-level distribution, health-score profile, and equipment-category information. These charts allow users to understand the condition of the simulated fleet without checking every machine individually. The health-score chart represents the condition of different machines and should not be interpreted as a time-series trend.

The recent alerts section highlights machines that require attention, especially those classified as Medium Risk, High Risk, or Critical. This helps users identify which machines may require observation, inspection, urgent checking, or immediate maintenance.

The machine-status table provides important information such as machine ID, equipment category, health score, alert level, maintenance priority, and schedule status. Users can select a machine from the dashboard and continue to the Machine Monitoring section to view more detailed operating data and maintenance information. the Dashboard interface presents machine conditions and maintenance information in a clear and simple form. It supports faster monitoring and helps users identify machines that require maintenance attentio

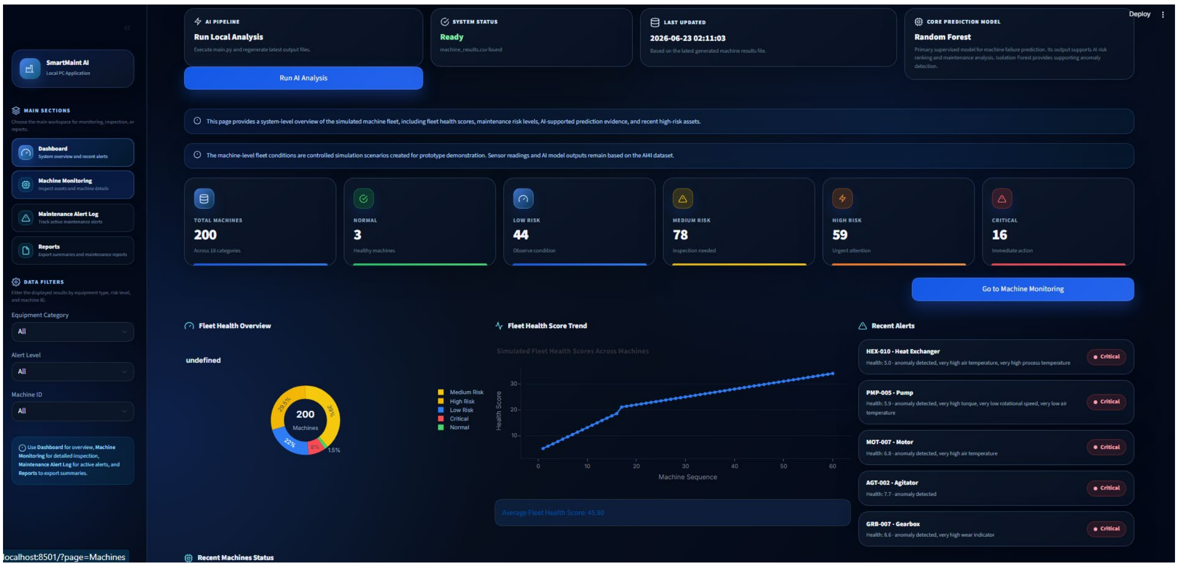


Figure 4.3: Dashboard Interface of SmartMaint AI

4.5.2 Machine monitoring interface

Figure 4.4 shows the Machine Monitoring interface of the SmartMaint AI system. This page allows users to inspect the condition of individual machines in more detail. It presents the monitored machines in a structured and easy-to-read form.

At the top of the interface, the system displays general information such as the analysis status and last updated time. Random Forest is used as the main supervised model for machine failure prediction, while Isolation Forest provides supporting anomaly evidence and One-Class SVM is used as an unsupervised comparison model.

The main section of the page displays the controlled simulated fleet in a table. The table contains important information such as machine ID, equipment category, health score, alert level, maintenance priority, and machine condition. The health score is shown as a numerical value and may also be supported by a visual indicator to make the condition easier to understand.

The alert-level column shows whether the machine is classified as Normal, Low Risk, Medium Risk, High Risk, or Critical. This helps users quickly identify which machines are healthy and which machines require observation, inspection, urgent attention, or immediate maintenance.

The filter section allows users to narrow the displayed results based on equipment category, alert level, or machine ID. This makes it easier to search for a specific machine or focus on machines with higher maintenance risk.

Users can select a machine to open its detailed information. The selected machine view displays the sensor readings, health score, alert level, detected problem, probable cause, observed abnormal pattern, recommended action, maintenance priority, and maintenance schedule suggestion.

The interface also allows users to export the filtered machine-monitoring results as a CSV file. This function is useful for documentation, record keeping, and further analysis outside the dashboard.

Figure 4.4: Machine Monitoring Interface of SmartMaint AI

4.5.3 Maintenance alert log interface

Figure shows the Maintenance Alert Log interface of the SmartMaint AI system. This page is designed to convert AI risk results into practical maintenance alerts that can be used by users to inspect, review, and plan maintenance actions. The alert log is loaded from the generated maintenance alert output file and displays machines that require attention based on their risk level and health score.

The overview section of the Maintenance Alert Log page. This section displays important summary cards such as total alerts, active alerts, critical alerts, high risk alerts, and average health score. These cards allow users to quickly understand the current maintenance condition of the system. The page also includes charts that show alerts by risk level and alerts by equipment category. These charts help users identify which risk levels and equipment types have the highest number of alerts.

Figure 4.5: Maintenance Alert Log Overview

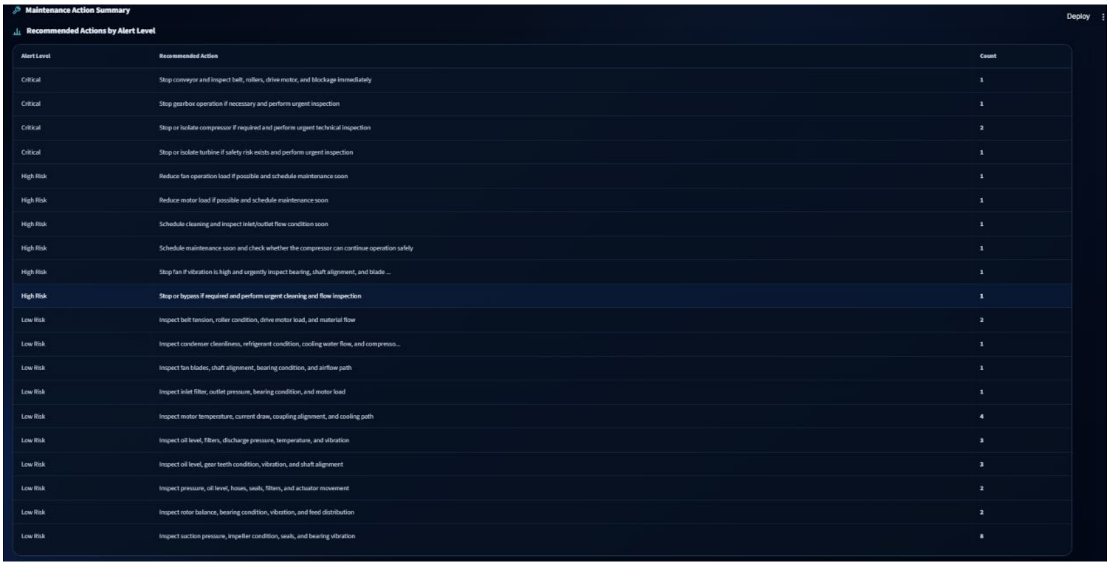
Figure shows the detailed Maintenance Alert Log table. The table contains important alert information such as alert ID, machine ID, equipment category, alert level, alert priority, health score, detected problem, probable cause, recommended action, observed pattern, alert status, and generated time. This table is important because it gives users complete maintenance information for each machine. Instead of only showing that a machine is abnormal, the system explains the problem, possible cause, and recommended action.

Maintenance Alert Log Table

Alert Id	Machine Id	Equipment Category	Alert Level	Alert Priority	Health Score	Problem	Probable Cause	Recommended Action	Observed Pattern	Alert Status	Generated At
ALT-0001	CNV-010	Conveyor	Critical	Immediate	54.37	Critical conveyor condition	Severe abnormal pattern may indicate belt jam, roller failure, or unsafe drive motor st...	Stop conveyor and inspect belt, rollers, drive motor, and blockage immediately	anomaly detected, very low conveyor drive load, very high conveyor speed variation	Active	2026-06-07 21:30:30
ALT-0002	TRB-002	Turbine	Critical	Immediate	55.84	Critical turbine condition	Severe abnormal pattern may indicate unsafe speed instability, thermal stress, or blade...	Stop or isolate turbine if safety risk exists and perform urgent inspection	anomaly detected, very low torque, high tool wear, very high rotational speed	Active	2026-06-07 21:30:30
ALT-0003	CMP-002	Compressor	Critical	Immediate	57.07	Critical compressor condition	Severe abnormal pattern may indicate major pressure instability, overheating, or lubric...	Stop or isolate compressor if required and perform urgent technical inspection	anomaly detected, very low torque, high tool wear, very high rotational speed	Active	2026-06-07 21:30:30
ALT-0004	GRB-004	Gearbox	Critical	Immediate	57.38	Critical gearbox condition	Severe abnormal pattern may indicate gear damage, lubrication failure, or unsafe torque...	Stop gearbox operation if necessary and perform urgent inspection	anomaly detected, very low mechanical load, very high shaft speed, high operating tempe...	Active	2026-06-07 21:30:30
ALT-0005	CMP-003	Compressor	Critical	Immediate	57.47	Critical compressor condition	Severe abnormal pattern may indicate major pressure instability, overheating, or lubric...	Stop or isolate compressor if required and perform urgent technical inspection	anomaly detected, very high torque, low rotational speed, very high air temperature	Active	2026-06-07 21:30:30
ALT-0006	FAN-007	Fan	High Risk	Urgent	57.9	Critical fan operating condition	Severe abnormal pattern may indicate bearing failure, blade damage, or unsafe vibration...	Stop fan if vibration is high and urgently inspect bearing, shaft alignment, and blade ...	anomaly detected, very high rotational airflow speed, very low airflow load demand, ver...	Active	2026-06-07 21:30:30
ALT-0007	HEX-009	Heat Exchanger	High Risk	Urgent	58.22	Critical heat exchanger condition	Severe abnormal pattern may indicate major blockage or thermal performance failure	Stop or bypass if required and perform urgent cleaning and flow inspection	anomaly detected, very high air temperature, high process temperature	Active	2026-06-07 21:30:30
ALT-0008	HEX-002	Heat Exchanger	High Risk	Urgent	59.74	High heat exchanger efficiency loss	Low health score may indicate fouling, blocked flow, or abnormal temperature difference	Schedule cleaning and inspect inlet/outlet flow condition soon	anomaly detected, low process temperature	Active	2026-06-07 21:30:30
ALT-0009	MOT-019	Motor	High Risk	Urgent	60.35	High motor stress condition	Low health score may indicate overload, insulation stress, or abnormal torque demand	Reduce motor load if possible and schedule maintenance soon	anomaly detected, very low torque, very high rotational speed, very high air temperature	Active	2026-06-07 21:30:30
ALT-0010	FAN-005	Fan	High Risk	Urgent	60.72	Severe fan airflow imbalance or bearing stress	Low health score may indicate high mechanical stress, bearing wear, or unstable rotation	Reduce fan operation load if possible and schedule maintenance soon	anomaly detected, very high rotational airflow speed, very low airflow load demand	Active	2026-06-07 21:30:30
ALT-0011	CMP-004	Compressor	High Risk	Urgent	61.17	High compressor stress condition	Low health score may indicate overheating, overpressure, or lubrication problem	Schedule maintenance soon and check whether the compressor can continue operation safely	anomaly detected, very low torque, very high rotational speed, very high air temperature	Active	2026-06-07 21:30:30
ALT-0012	HYD-001	Hydraulic System	Medium Risk	Inspect	55.59	Critical hydraulic system condition	Severe abnormal pattern may indicate major leakage, pump failure, or unsafe pressure co...	Stop hydraulic operation if unsafe and urgently inspect hoses, seals, pressure, and pump	anomaly detected, very low hydraulic load response, high process temperature	Active	2026-06-07 21:30:30
ALT-0013	FAN-012	Fan	Medium Risk	Inspect	57.33	Critical fan operating condition	Severe abnormal pattern may indicate bearing failure, blade damage, or unsafe vibration...	Stop fan if vibration is high and urgently inspect bearing, shaft alignment, and blade ...	anomaly detected, very high rotational airflow speed, very low airflow load demand, ver...	Active	2026-06-07 21:30:30
ALT-0014	CHL-009	Chiller	Medium Risk	Inspect	57.52	Critical chiller condition	Severe abnormal pattern may indicate major cooling failure, compressor stress, or therm...	Reduce cooling load or isolate chiller if required and perform urgent inspection	anomaly detected, very high cooling-side temperature, very high process cooling tempera...	Active	2026-06-07 21:30:30
ALT-0015	AGT-001	Agitator	Medium Risk	Inspect	58.15	Critical agitator condition	Severe abnormal pattern may indicate shaft damage, blade jam, or unsafe drive stress	Stop agitator if unsafe and urgently inspect shaft, blades, coupling, and motor	anomaly detected, very low mixing load demand, very high shaft speed variation	Active	2026-06-07 21:30:30

Figure 4.6: Maintenance Alert Log Table

Figure shows the Maintenance Action Summary section. This section summarizes recommended actions based on alert level and counts how many machines require each type of action. This helps users understand common maintenance actions that need to be taken, such as inspection, cleaning, checking mechanical load, or stopping equipment for urgent inspection. Overall, the Maintenance Alert Log interface supports maintenance planning by helping users prioritize machines according to risk level and recommended action.



Alert Level	Recommended Action	Count
Critical	Stop conveyor and inspect belt, rollers, drive motor, and blockage immediately	1
Critical	Stop gearbox operation if necessary and perform urgent inspection	1
Critical	Stop or isolate compressor if required and perform urgent technical inspection	2
Critical	Stop or isolate turbine if safety risk exists and perform urgent inspection	1
High Risk	Reduce fan operation load if possible and schedule maintenance soon	1
High Risk	Reduce motor load if possible and schedule maintenance soon	1
High Risk	Schedule cleaning and inspect inlet/outlet flow condition soon	1
High Risk	Schedule maintenance soon and check whether the compressor can continue operation safely	1
High Risk	Stop fan if vibration is high and urgently inspect bearing, shaft alignment, and blade ...	1
High Risk	Stop or bypass if required and perform urgent cleaning and flow inspection	1
Low Risk	Inspect belt tension, roller condition, drive motor load, and material flow	2
Low Risk	Inspect condenser cleanliness, refrigerant condition, cooling water flow, and compresso...	1
Low Risk	Inspect fan blades, shaft alignment, bearing condition, and airflow path	1
Low Risk	Inspect inlet filter, outlet pressure, bearing condition, and motor load	1
Low Risk	Inspect motor temperature, current draw, coupling alignment, and cooling path	4
Low Risk	Inspect oil level, filters, discharge pressure, temperature, and vibration	3
Low Risk	Inspect oil level, gear teeth condition, vibration, and shaft alignment	3
Low Risk	Inspect pressure, oil level, hoses, seals, filters, and actuator movement	2
Low Risk	Inspect rotor balance, bearing condition, vibration, and feed distribution	2
Low Risk	Inspect suction pressure, impeller condition, seals, and bearing vibration	8

Figure 4.7: Maintenance Action Summary

4.5.4 Reports Interface

This figure shows the Reports interface of the SmartMaint AI system. This page is designed to help users export and review maintenance results generated by the system. The Reports page provides a summary of analysed machines, average health score, high risk machines, and critical machines. These summary cards allow users to quickly understand the overall maintenance condition before downloading the report. the Reports Centre overview. This section includes two main report download options, which are the filtered system report and the maintenance summary report. The filtered system report allows users to export the machine results based on the current filters, while the maintenance summary report provides a simpler text summary suitable for supervisor review, project demonstration, or documentation purposes.

The Reports page also includes visual summaries such as the alert level summary and top high risk machines. The alert level summary shows the number of machines classified as Normal, Low Risk, Medium Risk, High Risk, and Critical. The top high risk machines chart helps users identify machines with lower health scores that may require earlier inspection or maintenance action.

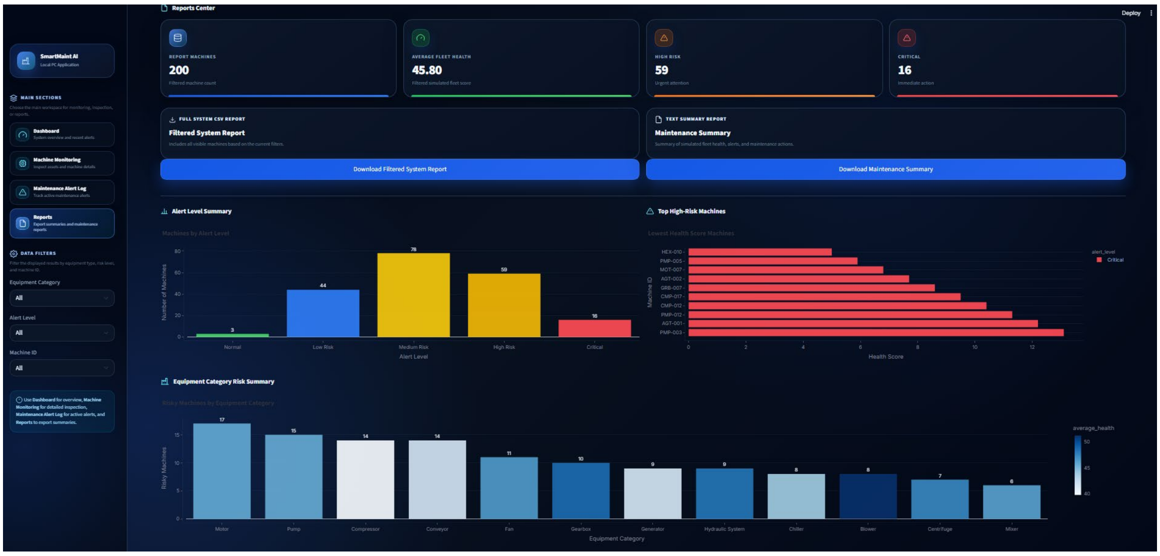


Figure 4.8: Reports Center Overview

The equipment category risk summary and report preview table. The equipment category risk summary shows which equipment categories have more risky machines, such as motors, pumps, compressors, and heat exchangers. This helps users understand which types of equipment require more attention. The report preview table displays machine ID, equipment type, health score, status, last anomaly, and action button. This allows users to check the report content before exporting it. the Reports interface supports documentation, maintenance review, and decision making by converting AI results into downloadable and readable maintenance reports.

4.5.5 Selected Machine Details

shows the Selected Machine Details interface of the SmartMaint AI system. This page appears when the user selects a specific machine from the Machine Monitoring table. Its purpose is to provide a clearer and more detailed explanation of the selected machine condition.

In this example, the selected machine is PMP-016, which belongs to the Pump equipment category. The system displays important information such as the machine ID, equipment category, health score, alert level, maintenance priority, and supporting anomaly result.

The selected machine is classified as High Risk with an Urgent maintenance priority. This condition is part of the controlled simulated fleet used to demonstrate the monitoring and maintenance decision-support functions of the system.

The interface also provides maintenance information, including the detected problem, probable cause, observed abnormal pattern, recommended action, and AI explanation. For this machine, the detected problem is a high-risk pump operating condition.

The probable cause section presents possible reasons such as cavitation, bearing wear, or unstable hydraulic load. These causes are suggestions generated from the machine condition, equipment category, and available sensor patterns. They should be treated as possible inspection areas rather than confirmed causes of failure.

The observed abnormal pattern shows that the machine has very high air temperature and process temperature readings. This information helps users understand which operating values may have contributed to the maintenance warning.

Based on the machine condition, the system recommends planning maintenance soon and reducing the pump load where possible. The interface may also provide a maintenance schedule suggestion to support the next inspection or maintenance activity.

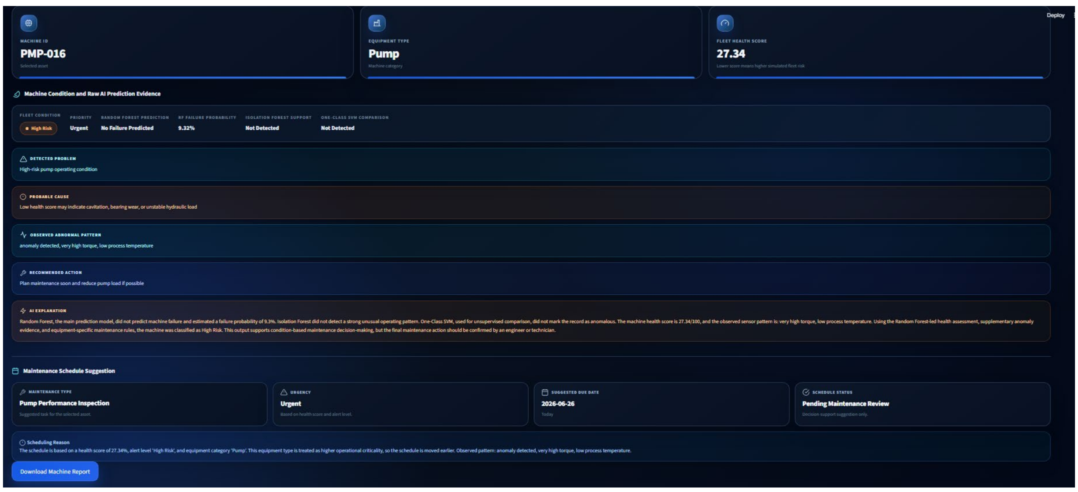


Figure 4.9: Selected Machine Details Interface

Shows the Maintenance Schedule Suggestion and Sensor Readings sections for the selected machine. The system provides a suggested maintenance activity, urgency level, schedule status, due date, and reason based on the machine condition.

In this example, the system suggests a **Pump Performance Inspection** with **Routine** urgency and a **Pending Maintenance Review** status. The suggested due date is prepared using rule-based decision logic that considers the machine health score, alert level, equipment category, and observed abnormal pattern. This helps users plan suitable maintenance actions instead of only viewing the machine risk result. The Sensor Readings section displays the machine operating values used in the analysis. These readings include machine type, air temperature, process temperature, rotational speed, torque, and tool wear. Showing the original sensor readings together with the machine condition helps users understand the operating data related to the generated maintenance information. The interface also connects the machine health score and alert level with the detected problem, observed abnormal pattern, recommended action, and maintenance schedule suggestion. However, the simulated health score and alert level are used for prototype demonstration and should not be treated as direct Random Forest failure probabilities.

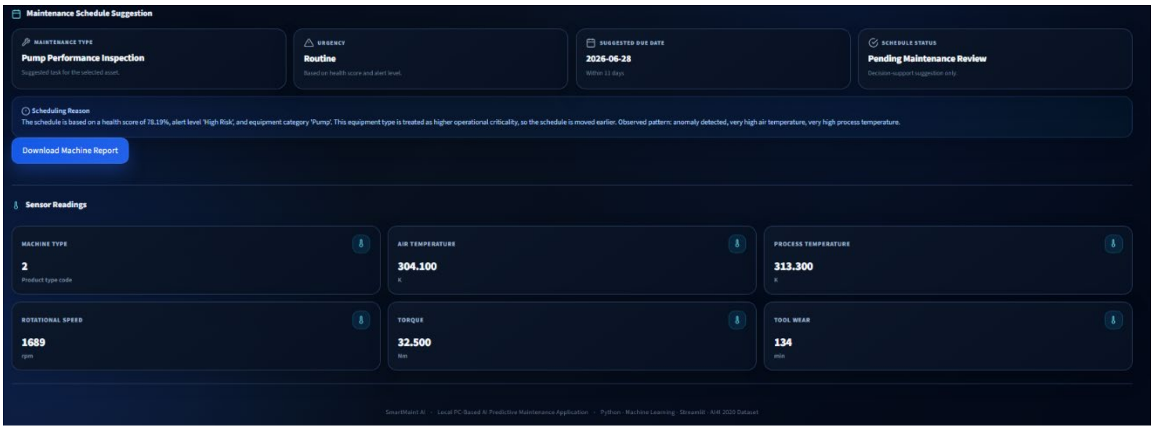


Figure 4.10: Maintenance Schedule Suggestion and Sensor Readings

4.6 Conclusion

This chapter explained the design of the SmartMaint AI system. It discussed the system architecture, system flow, module design, and user interface design used to organise the system and connect its main functions.

The system architecture showed how data from the AI4I 2020 Predictive Maintenance dataset moves through data loading, preprocessing, machine learning analysis, model evaluation, controlled fleet simulation, decision support, output generation, and dashboard presentation.

Random Forest is used as the main supervised model for machine failure prediction. Isolation Forest provides supporting anomaly detection evidence, while One-Class SVM is used as an unsupervised comparison model. The original sensor readings and model outputs are kept as AI evidence.

The system flow explained the step-by-step process from loading and preparing the dataset to generating the final dashboard, maintenance alert log, maintenance schedule suggestions, and reports. It also included the controlled simulation of 200 machines from 18 equipment categories to support machine-level monitoring and demonstrate all five alert conditions.

The module design described the main components of the system, including the Dashboard Module, Machine Monitoring Module, AI Prediction and Health Scoring Module, Maintenance Alert Log Module, and Report Generation Module. Each module performs a specific function and works together with the other modules.

The user interface design showed how the Streamlit dashboard presents machine conditions and maintenance information in a simple and understandable way. Users can view the overall fleet condition, monitor individual machines, review sensor readings, check maintenance alerts, view recommendations and schedule suggestions, and generate reports. the SmartMaint AI design supports the development of a practical local PC-based predictive maintenance application. It connects machine operating data, machine learning evidence, controlled simulated conditions, and maintenance decision support to help users understand machine health and plan suitable maintenance actions.

CHAPTER 5: IMPLEMENTATION

5.1 Introduction

This chapter explains the implementation of the SmartMaint AI system based on the requirements and system design discussed in the previous chapters. It describes how the proposed design was developed into a working local PC-based predictive maintenance application.

The main implementation activities include dataset loading, data preprocessing, machine learning model development, model evaluation, controlled fleet simulation, machine health score preparation, alert-level classification, maintenance decision support, dashboard development, maintenance alert logging, scheduling, and report generation.

SmartMaint AI was developed using Python and Streamlit. Python was used for data processing, machine learning, evaluation, system logic, and report generation. Streamlit was used to create the local dashboard interface so that users can view machine conditions and maintenance information clearly.

The AI4I 2020 Predictive Maintenance dataset was used as the main source of machine operating data. The system analyses features such as machine type, air temperature, process temperature, rotational speed, torque, and tool wear. The Machine failure column is used as the labelled target for training and evaluating the Random Forest model.

Random Forest was implemented as the main supervised model for machine failure prediction because it achieved the best overall evaluation performance. It produces a failure prediction and failure probability for each analysed record. Isolation Forest was implemented as a supporting unsupervised anomaly detection model, while One-Class SVM was used as an unsupervised comparison model.

The original sensor readings and machine learning outputs are kept as AI evidence. Since the dataset does not provide complete machine histories or an actual industrial fleet, a controlled simulated fleet of 200 machines from 18 equipment categories was implemented for prototype demonstration. This allows the system to display machine-level health scores and classify machines as Normal, Low Risk, Medium Risk, High Risk, or Critical.

The implementation was divided into separate modules so that each system function could be developed, tested, and maintained more easily. The decision-support component generates information such as the detected problem, probable cause, observed abnormal pattern, recommended action, maintenance priority, and maintenance schedule suggestion.

The results are presented through the Streamlit dashboard. Users can view the overall fleet condition, monitor individual machines, review sensor readings, check active maintenance alerts, inspect schedule suggestions, and download machine or system summary reports.

5.2 Development Environment and Tools

The SmartMaint AI system was developed in a local computing environment using the Windows operating system. This environment supported source code development, dataset processing, machine learning model training, system execution, and the display of the Streamlit dashboard.

A cloud platform was not required because all data processing, model execution, machine simulation, and system analysis were carried out locally on a personal computer. This made the system suitable for local demonstration without depending on an external server.

Python was used as the main programming language because it provides many useful libraries for data processing, machine learning, visualisation, and application development. Visual Studio Code was used to write, organise, and manage the project source code and supporting files.

Streamlit was selected to develop the user interface because it allows Python results to be displayed through an interactive web-based dashboard running on the local computer. Users can access the dashboard through a supported web browser while the Streamlit application is active.

Several Python libraries were used during the implementation. Pandas was used to load, clean, transform, and manage the dataset. NumPy supported numerical calculations and array operations. Scikit-learn was used to implement the Random Forest, Isolation Forest, and One-Class SVM models. It was also used for feature standardisation, dataset splitting, and model performance evaluation.

Random Forest was implemented as the main supervised machine failure prediction model. Isolation Forest was used to provide supporting anomaly detection evidence, while One-Class SVM was used as an unsupervised comparison model. Plotly was used to create interactive charts and visual summaries for the Streamlit dashboard.

The project was organised into separate components for data loading, preprocessing, model training, evaluation, decision support, explainability, machine simulation, maintenance alert management, and dashboard presentation. This modular structure made the system easier to understand, test, maintain, and improve.

The generated outputs, including model results, simulated machine information, maintenance alerts, evaluation results, and maintenance recommendations, were stored in CSV files. These files were then loaded into the Streamlit application for dashboard presentation and report generation.

HTML and CSS were also used within the Streamlit application to improve the layout and visual appearance of the dashboard. HTML was used to organise some interface elements, while CSS was used to customise colours, cards, buttons, spacing, fonts, borders, and alert indicators.

Overall, the selected development environment and software tools supported the complete implementation of SmartMaint AI as a local PC-based predictive maintenance application.

Figure 5.1: Visual Studio Code

Figure 5.2: Streamlit

Figure 5.3: Python

Figure 5.4: HTML

Figure 5.5: Scikit learn

Figure 5.6: Plotly

Table 5.1: SmartMaint AI Development Environment

Software or Library	Purpose
Windows Operating System	Used as the main operating system to develop and run the SmartMaint AI application.
Python	Used as the main programming language for data processing, machine learning, system logic, and report generation.
Visual Studio Code	Used to write, organise, and manage the project source code.
Streamlit & HTML	Used to develop the interactive dashboard and user interface.
Scikit learn	Used for data scaling, model training, anomaly detection, classification, and model evaluation.
Plotly	Used to create interactive charts and visual summaries in the dashboard.

5.3 Dataset Loading and Preparation

The AI4I 2020 Predictive Maintenance dataset was used as the main dataset for developing SmartMaint AI. It is a synthetic industrial predictive maintenance dataset containing 10,000 machine operating records. The records represent normal operating conditions and different machine failure conditions. The dataset was stored in CSV format and loaded into the system using the Pandas library.

The main input features selected from the dataset were machine type, air temperature, process temperature, rotational speed, torque, and tool wear.

These features describe different parts of the machine operating condition and are suitable for machine failure prediction and anomaly detection. The Machine failure column was used as the labelled target for training and evaluating the Random Forest model.

To keep the project organised, the dataset-loading function was placed in a separate Python module. When the system starts, the function reads the CSV file from the project directory and converts it into a Pandas DataFrame. The function also checks whether the file exists before loading it. This helps prevent the program from stopping unexpectedly when the dataset is missing or the file path is incorrect.

Before preprocessing, the dataset was examined to understand its structure. The inspection included checking the number of rows and columns, column names, data types, missing values, duplicate records, and the distribution of the Machine failure target.

The dataset contains more normal records than failure records. This means that the target classes are imbalanced. The imbalance is important during model evaluation because accuracy alone may not provide a complete explanation of model performance. Therefore, additional measures such as precision, recall, F1-score, ROC-AUC, and PR-AUC were also used.

Identification columns such as UDI and Product ID were excluded from the machine learning input because they identify individual records but do not directly describe the machine operating condition. The remaining features were passed to the preprocessing component.

After the dataset was loaded successfully, the prepared operating features were used by the three machine learning models. Random Forest used the Machine failure label for supervised machine failure prediction. Isolation Forest analysed the operating features to produce supporting anomaly evidence, while One-Class SVM was used as an unsupervised comparison model.

The original records and model outputs were also used to support the controlled simulation of 200 machines from 18 equipment categories. The simulated fleet was then used for machine-level monitoring, health score presentation, alert-level classification, maintenance decision support, and report generation.

Table 5.2: Dataset Features

Feature	Description
Type	Product quality category represented as Low, Medium, or High
Air Temperature	Temperature of the surrounding air measured in Kelvin
Process Temperature	Temperature generated during the machine process measured in Kelvin
Rotational Speed	Machine rotation speed measured in revolutions per minute (rpm)
Torque	Mechanical force applied during machine operation measured in Newton metres (Nm)
Tool Wear	Total tool usage or wear time measured in minutes
Machine Failure	Binary label indicating whether a machine failure occurred

5.4 Data Preprocessing

Before implementing the machine learning models, the dataset was preprocessed to ensure that the input data was suitable for analysis. This step was necessary because the dataset contains categorical data and numerical features measured using different units and value ranges.

The first preprocessing step involved selecting the required features from the AI4I 2020 Predictive Maintenance dataset. The selected input features were machine type, air temperature, process temperature, rotational speed, torque, and tool wear. Identification columns such as UDI and Product ID were removed because they only identify individual records and do not describe the operating condition of the machine.

The Machine type feature contains the categorical values L, M, and H, which represent low, medium, and high product-quality categories. Since machine learning models require numerical input, these values were converted into numerical form. In the system, L was encoded as 0, M as 1, and H as 2.

The numerical features were prepared using StandardScaler from Scikit-learn. Standardisation changes the features to a similar scale while keeping the relationships between their values. This step is especially important for One-Class SVM because its results can be affected by differences in feature scale. It also provides consistent prepared input for the anomaly detection models.

For supervised Random Forest training and evaluation, the Machine failure column was used as the target label. The selected operating features were used as the input data, while the target indicated whether a machine failure occurred.

Random Forest then learned the relationship between the operating conditions and the known failure labels.

Isolation Forest and One-Class SVM used the prepared machine operating features without using the Machine failure column during anomaly detection. Isolation Forest generated supporting anomaly labels and scores, while One-Class SVM was used as an unsupervised comparison model.

The prepared dataset was then divided into training and testing data where required for supervised model development. After preprocessing was completed, the processed features were passed to the machine learning implementation and evaluation stages.

```python
columns_to_drop = ["UDI", "Product ID"]

for col in columns_to_drop:
    if col in df.columns:
        df.drop(columns=col, inplace=True)

# Encode Type column: L, M, H
if "Type" in df.columns:
    type_mapping = {
        "L": 0,
        "M": 1,
        "H": 2
    }
    df["Type"] = df["Type"].map(type_mapping)

# Input features
feature_columns = [
    "Type",
    "Air temperature [K]",
    "Process temperature [K]",
    "Rotational speed [rpm]",
    "Torque [Nm]",
    "Tool wear [min]"
]

# Target columns for evaluation
target_columns = [
    "Machine failure",
    "TWF",
    "HDF",
    "PWF",
    "OSF",
    "RNF"
]
```

Figure 5.7: Feature Selection and Categorical Encoding

```python
df.dropna(inplace=True)

# Separate features and targets
X = df[feature_columns].copy()
y = df[target_columns].copy()

# Scale the input features
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

X_scaled_df = pd.DataFrame(X_scaled, columns=feature_columns)

# Data before scaling
cleaned_data = pd.concat(
    [X.reset_index(drop=True), y.reset_index(drop=True)],
    axis=1
)

# Data after scaling
scaled_data = pd.concat(
    [X_scaled_df.reset_index(drop=True), y.reset_index(drop=True)],
    axis=1
)

print("\nPreprocessing completed successfully.")
print(f"Cleaned data shape: {cleaned_data.shape}")
print(f"Scaled data shape: {scaled_data.shape}")

return cleaned_data, scaled_data, scaler
```

Figure 5.8: Data Cleaning and Numerical Feature Scaling

5.5 Machine Learning Model Implementation

The machine learning component receives the preprocessed machine operating data and applies three models: Random Forest, Isolation Forest, and One-Class SVM. These models were implemented using the Scikit-learn library.

Random Forest was implemented as the main supervised machine learning model for machine failure prediction. The selected machine operating features were used as the input variables, while the Machine failure column was used as the target label. Before training, the dataset was divided into training and testing subsets so that the model could be evaluated using unseen data.

After training, Random Forest produced a predicted failure class and failure probability for each testing record. It was selected as the main model because it achieved the best overall performance during evaluation. The model was assessed using accuracy, precision, recall, F1-score, ROC-AUC, and PR-AUC.

Random Forest also produced feature importance values. These values show which input features had a greater influence on the prediction result. Features such as torque, rotational speed, tool wear, air temperature, process temperature, and machine type were included in this analysis. However, feature importance does not prove that a feature directly caused the failure.

Isolation Forest was implemented as a supporting unsupervised anomaly detection model. It analysed the standardized machine operating features without using the Machine failure label during anomaly detection. The model produced an anomaly label and anomaly score for each record. These results were kept as supporting evidence of unusual machine behaviour.

One-Class SVM was implemented as an unsupervised comparison model. It used the same standardized input features and classified each record as normal or abnormal. Its results were mainly used to compare another anomaly detection approach with Isolation Forest.

After the three models completed their predictions, the outputs were saved for evaluation, explainability, and dashboard presentation. The original sensor readings and model outputs were kept as AI evidence.

The final machine-level health scores and alert levels shown in the dashboard were prepared through the controlled simulated fleet used for prototype demonstration. Therefore, they were not calculated directly from the Random Forest failure probability or the Isolation Forest anomaly score.

Overall, the machine learning component provided failure prediction, anomaly evidence, model comparison, and feature-importance information to support the remaining SmartMaint AI functions.

```
model = IsolationForest(
    n_estimators=200,
    contamination=ANOMALY_RATE,
    random_state=RANDOM_STATE,
    n_jobs=-1,
)

model.fit(normal_train_X)

# Evaluation on the held-out test set
test_predictions = model.predict(X_test)
test_anomaly_labels = [1 if pred == -1 else 0 for pred in test_predictions]

# decision_function: higher score = more normal
# multiply by -1 to create operational risk score
test_normality_scores = model.decision_function(X_test)
test_risk_scores = -test_normality_scores

metrics = evaluate_model_predictions(
    "Isolation Forest",
    y_test,
    test_anomaly_labels,
    y_score=test_risk_scores,
)

# Final monitoring predictions for the dashboard output
X_full = scaled_data[feature_columns]
full_predictions = model.predict(X_full)
full_normality_scores = model.decision_function(X_full)
full_anomaly_labels = [1 if pred == -1 else 0 for pred in full_predictions]

result = scaled_data.copy()
result["if_anomaly_label"] = full_anomaly_labels
result["if_anomaly_score"] = full_normality_scores
result["if_risk_score"] = -full_normality_scores
```

Figure 5.9: Isolation Forest Implementation

```python
model = OneClassSVM(
    kernel="rbf",
    gamma="scale",
    nu=ANOMALY_RATE,
)

model.fit(normal_train_X)

# Evaluation on the held-out test set
test_predictions = model.predict(X_test)
test_anomaly_labels = [1 if pred == -1 else 0 for pred in test_predictions]

# decision_function: higher score = more normal
# multiply by -1 to create operational risk score
test_normality_scores = model.decision_function(X_test)
test_risk_scores = -test_normality_scores

metrics = evaluate_model_predictions(
    "One-Class SVM",
    y_test,
    test_anomaly_labels,
    y_score=test_risk_scores,
)

# Final monitoring predictions for the dashboard output
X_full = scaled_data[feature_columns]
full_predictions = model.predict(X_full)
full_normality_scores = model.decision_function(X_full)
full_anomaly_labels = [1 if pred == -1 else 0 for pred in full_predictions]

result = scaled_data.copy()
result["ocsvm_anomaly_label"] = full_anomaly_labels
result["ocsvm_anomaly_score"] = full_normality_scores
result["ocsvm_risk_score"] = -full_normality_scores
```

Figure 5.10: One-Class SVM Implementation

```python
model = RandomForestClassifier(
    n_estimators=300,
    random_state=RANDOM_STATE,
    class_weight="balanced_subsample",
    min_samples_leaf=2,
    n_jobs=-1,
    bootstrap=True,
    oob_score=True,
)

model.fit(X_train, y_train)

# Evaluation on the held-out test set
y_pred = model.predict(X_test)
y_proba = model.predict_proba(X_test)[:, 1]

metrics = evaluate_model_predictions(
    "Random Forest",
    y_test,
    y_pred,
    y_score=y_proba,
)

metrics["oob_score"] = getattr(model, "oob_score_", None)

test_results = X_test.copy()
test_results["actual_failure"] = y_test.values
test_results["rf_prediction"] = y_pred
test_results["rf_failure_probability"] = y_proba

# Final monitoring predictions for every machine/record
X_full = scaled_data[feature_columns]
full_predictions = model.predict(X_full)
full_probabilities = model.predict_proba(X_full)[:, 1]

result = scaled_data.copy()
result["rf_prediction"] = full_predictions
result["rf_failure_probability"] = full_probabilities

feature_importance = pd.DataFrame(
```

Figure 5.11: Random Forest Implementation

5.6 Health Score and Alert Level Classification

The machine learning results and machine operating data are presented in a simpler form so that users can understand the machine condition more easily. Instead of displaying only technical outputs such as a failure probability, anomaly label, or anomaly score, SmartMaint AI also presents each simulated machine using a health score ranging from 0 to 100.

A higher health score represents a healthier machine condition, while a lower health score shows that the machine may require more maintenance attention. The system uses five health score ranges:

- **Normal:** 80 to 100
- **Low Risk:** 60 to below 80
- **Medium Risk:** 40 to below 60
- **High Risk:** 20 to below 40
- **Critical:** Below 20

The original sensor readings and model results are kept as AI evidence. Random Forest provides the main machine failure prediction, Isolation Forest provides supporting anomaly evidence, and One-Class SVM is used as an unsupervised comparison model.

However, the final machine-level health scores and alert levels displayed in the dashboard are prepared through the controlled simulated fleet. This simulation was used because the AI4I 2020 Predictive Maintenance dataset does not provide complete machine histories, real equipment identities, or an actual industrial fleet.

The controlled simulation contains 200 machines from 18 equipment categories. It allows the system to demonstrate all five alert levels and test the machine-monitoring, maintenance-alert, recommendation, scheduling, and report-generation functions.

Each alert level is connected to a maintenance priority. Machines classified as Normal can continue operating under routine monitoring. Low Risk machines may require observation, Medium Risk machines may require inspection, High Risk machines are given urgent attention, and Critical machines require immediate action.

The health score, alert level, maintenance priority, and supporting anomaly information are stored with the processed machine results. These values are displayed in the dashboard, Maintenance Alert Log, selected machine details, maintenance schedule suggestions, and generated reports.

The final simulated health score should not be interpreted as a direct Random Forest failure probability or as a value calculated only from the Isolation Forest anomaly score. It is a machine condition value used for prototype demonstration and maintenance decision support.

Table 5.3: Machine Alert Levels and Maintenance Priority

Alert Level	General Meaning	Maintenance Priority
Normal	The machine is operating under a healthy condition	Regular monitoring
Low Risk	A small abnormal condition may be present	Observe
Medium Risk	The machine condition requires inspection	Inspect
High Risk	The machine shows a serious abnormal condition	Urgent
Critical	The machine may require immediate maintenance action	Immediate

Type	Air temperature [K]	Process temperature [K]	Rotational speed [rpm]	Torque [Nm]	Tool wear [min]	Machine failure	TWF	HDF	PWF	OSF	RNF	if_anomaly_label	if_anomaly_score	if_risk_score	ocsvm_anomaly_label	ocsvm_anomaly_score	ocsvm_risk_score
0.744413	-0.952389443	-0.947359888	0.06818514	0.282199756	-1.695983744	0	0	0	0	0	0	0	0.098677706	-0.09867771	0	3.467848427	-3.467848427
-0.74531	-0.90239341	-0.879959002	-0.729471505	0.63330802	-1.648851701	0	0	0	0	0	0	0	0.118107646	-0.11810765	0	1.558179667	-1.558179667
-0.74531	-0.952389443	-1.014760773	-0.22744984	0.944289625	-1.617430338	0	0	0	0	0	0	0	0.114057041	-0.11405704	0	1.394777277	-1.394777277
-0.74531	-0.90239341	-0.947359888	-0.590021043	-0.04884518	-1.586008976	0	0	0	0	0	0	0	0.124641052	-0.12464105	0	0.972877804	-0.972877804
-0.74531	-0.90239341	-0.879959002	-0.729471505	0.001313145	-1.554587614	0	0	0	0	0	0	0	0.128820615	-0.12882061	0	0.86129964	-0.86129964
0.744413	-0.952389443	-0.947359888	-0.634645191	0.191914774	-1.523166251	0	0	0	0	0	0	0	0.119535106	-0.11953511	0	3.915812928	-3.915812928
-0.74531	-0.952389443	-0.947359888	0.10723127	0.242073097	-1.476034208	0	0	0	0	0	0	0	0.142364469	-0.14236447	0	1.942766961	-1.942766961
-0.74531	-0.952389443	-0.947359888	-0.065687304	0.021376474	-1.444612845	0	0	0	0	0	0	0	0.147347458	-0.14734746	0	2.120625978	-2.120625978
0.744413	-0.852397376	-0.879959002	0.715235286	-1.14229663	-1.413191483	0	0	0	0	0	0	0	0.100340015	-0.10034001	0	3.55982802	-3.55982802
0.744413	-0.752405309	-0.677756345	1.128008655	-1.20248662	-1.366059439	0	0	0	0	0	0	0	0.089467654	-0.08946765	0	3.822342591	-3.822342591
2.234132	-0.802401343	-0.745157231	1.356707414	-1.61378487	-1.318927396	0	0	0	0	0	0	0	0.008464307	-0.00846431	0	0.441151574	-0.441151574
2.234132	-0.702409275	-0.61035546	-0.645801228	0.432674726	-1.24037399	0	0	0	0	0	0	0	0.081359156	-0.08135916	0	2.188047641	-2.188047641
0.744413	-0.702409275	-0.61035546	-1.114354782	1.114827925	-1.161820584	0	0	0	0	0	0	0	0.107870368	-0.10787037	0	5.229023053	-5.229023053
0.744413	-0.702409275	-0.542954574	1.133586674	-1.00185332	-1.114688541	0	0	0	0	0	0	0	0.104337978	-0.10433798	0	4.68185808	-4.68185808
-0.74531	-0.702409275	-0.542954574	2.767946094	-2.04514645	-1.067556497	0	0	0	0	0	0	0	0.026919051	-0.02691905	0	1.608930617	-1.608930617
-0.74531	-0.702409275	-0.542954574	0.017982974	0.843972978	-1.036135135	0	0	0	0	0	0	0	0.146084042	-0.14608404	0	2.538876293	-2.538876293
0.744413	-0.702409275	-0.542954574	-1.2705393	0.663403014	-1.004713772	0	0	0	0	0	0	0	0.115950953	-0.11595095	0	4.556576778	-4.556576778
0.744413	-0.652413242	-0.542954574	-0.718315468	0.563086367	-0.957581729	0	0	0	0	0	0	0	0.148581643	-0.14858164	0	5.981416735	-5.981416735
2.234132	-0.602417208	-0.542954574	-1.298429392	1.455904524	-0.910449685	0	0	0	0	0	0	0	0.029971263	-0.02997126	0	1.753676062	-1.753676062
0.744413	-0.552421174	-0.475553688	0.520004639	-0.75106171	-0.831896279	0	0	0	0	0	0	0	0.147735163	-0.14773516	0	5.443029242	-5.443029242
2.234132	-0.552421174	-0.475553688	-0.913546116	0.272168091	-0.784764236	0	0	0	0	0	0	0	0.082123295	-0.0821233	0	2.314536379	-2.314536379
-0.74531	-0.602417208	-0.475553688	-0.495194728	0.48283305	-0.70621083	0	0	0	0	0	0	0	0.172550982	-0.17255098	0	3.882982129	-3.882982129
0.744413	-0.552421174	-0.475553688	0.235525695	-0.93163167	-0.674789467	0	0	0	0	0	0	0	0.138368409	-0.13836841	0	4.840823244	-4.840823244
-0.74531	-0.502425141	-0.408152803	1.22283497	-1.4332149	-0.627657424	0	0	0	0	0	0	0	0.11384858	-0.11384858	0	2.57758443	-2.57758443
0.744413	-0.502425141	-0.408152803	0.123965325	-0.2695418	-0.596236062	0	0	0	0	0	0	0	0.156412531	-0.15641253	0	6.11409561	-6.11409561
-0.74531	-0.502425141	-0.340751917	1.797370875	-1.67397486	-0.549104018	0	0	0	0	0	0	0	0.081353118	-0.08135312	0	2.713880465	-2.713880465
-0.74531	-0.452429107	-0.340751917	-0.149357581	-0.0990035	-0.517682656	0	0	0	0	0	0	0	0.176705476	-0.17670548	0	3.500078881	-3.500078881
2.234132	-0.452429107	-0.408152803	1.51846995	-1.54356322	-0.486261293	0	0	0	0	0	0	0	0.019648566	-0.01964857	0	2.478348417	-2.478348417
-0.74531	-0.452429107	-0.408152803	-0.556552932	0.422643062	-0.407707887	0	0	0	0	0	0	0	0.175072364	-0.17507236	0	4.079778563	-4.079778563
-0.74531	-0.502425141	-0.408152803	0.860263767	-0.99182166	-0.376286525	0	0	0	0	0	0	0	0.138988289	-0.13898829	0	3.078278995	-3.078278995
0.744413	-0.452429107	-0.340751917	-1.114354782	0.823909649	-0.344865163	0	0	0	0	0	0	0	0.131431109	-0.13143111	0	6.375390647	-6.375390647
-0.74531	-0.502425141	-0.408152803	1.44595571	-1.45327823	-0.297733119	0	0	0	0	0	0	0	0.10813226	-0.10813226	0	2.946067535	-2.946067535
-0.74531	-0.502425141	-0.408152803	-0.668113302	0.833941314	-0.266311757	0	0	0	0	0	0	0	0.168388261	-0.16838826	0	4.490260009	-4.490260009
-0.74531	-0.552421174	-0.475553688	0.704079249	-0.75106171	-0.234890394	0	0	0	0	0	0	0	0.14561645	-0.14561645	0	3.310541554	-3.310541554

Figure 5.12: Sample Anomaly Detection Output from Models

5.7 Maintenance Recommendation and Alert Log Generation

After the machine health score and alert level are prepared, the system generates maintenance information for each simulated machine. This step makes the machine condition easier for users to understand. Instead of showing only technical outputs such as a failure probability or anomaly score, SmartMaint AI provides practical maintenance information.

The generated information includes the detected problem, probable cause, observed abnormal pattern, recommended action, maintenance priority, and maintenance schedule suggestion. These outputs are produced using rule-based decision logic that considers the equipment category, health score, alert level, and available sensor patterns.

For example, when a pump has a low health score together with unusual temperature, torque, or rotational-speed readings, the system may identify a high-risk pump operating condition. It may then recommend inspecting the pump, checking for cavitation or bearing problems, and reducing the operating load where possible.

Different explanations and actions are generated for different equipment categories. This is important because motors, pumps, fans, compressors, gearboxes, and other rotating equipment may experience different operating problems. Equipment-specific rules help make the recommendations more relevant to the selected machine.

The system also assigns a maintenance priority based on the alert level. Low Risk machines are generally assigned the Observe priority, Medium Risk machines are assigned Inspect, High Risk machines receive Urgent priority, and Critical machines receive Immediate priority. Normal machines remain under routine monitoring.

A maintenance schedule suggestion is also prepared for each machine. It includes the maintenance type, urgency, schedule status, suggested due date, and reason. The due date and urgency depend on the machine health score, alert level, equipment category, and observed condition.

The generated maintenance information is stored in the Maintenance Alert Log. Each alert record may contain the alert ID, machine ID, equipment category, alert level, priority, health score, detected problem, observed abnormal pattern, recommended action, schedule status, suggested due date, alert status, and generation time.

The Maintenance Alert Log is displayed through the Streamlit dashboard. Users can review the machines that require attention and filter the results by alert level, equipment category, priority, or maintenance condition. This helps users identify which machines should be checked first.

The system also produces maintenance summaries by grouping machines according to alert level, equipment category, priority, or recommended action. These summaries help users understand the most common maintenance needs across the simulated fleet.

The generated problems, probable causes, and recommendations are decision-support suggestions. They do not confirm the actual physical cause of a machine problem and should be supported by technician inspection before maintenance work is carried out.

Table 5.4: Maintenance Alert Log Information

Field	Description
Alert ID	Unique identification number for each maintenance alert
Machine ID	Identification number of the affected machine
Equipment Category	Type of machine, such as pump, motor, fan, or compressor
Alert Level	Risk category assigned to the machine
Alert Priority	Maintenance urgency such as Observe, Inspect, Urgent, or Immediate
Health Score	Calculated condition score of the machine
Detected Problem	Main machine problem identified by the system
Probable Cause	Possible reason for the abnormal machine condition
Observed Pattern	Abnormal sensor pattern found during analysis
Recommended Action	Suggested maintenance action for the machine
Alert Status	Shows whether the maintenance alert is active
Generated Time	Date and time when the alert was created

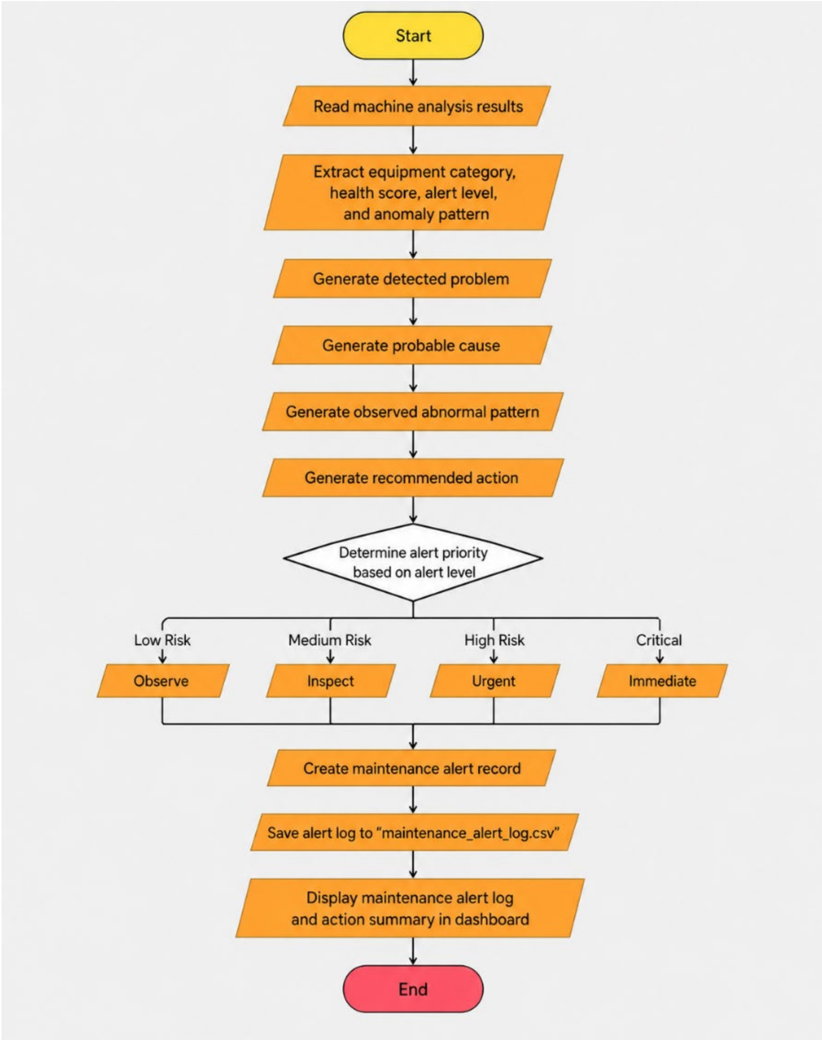


Figure 5.13: Maintenance Recommendation and Alert Log Implementation

5.8 Report Generation

The report generation function allows users to save and review the maintenance results produced by SmartMaint AI. This function supports documentation, maintenance planning, project demonstration, and further review of machine conditions outside the dashboard. On the Reports Centre page, users are provided with options to download system-level results. The filtered system report is generated in CSV format and contains the machine records currently displayed according to the selected filters.These filters may include equipment category, alert level, priority, or machine

ID. This allows users to export only the information that is relevant to their monitoring needs.

The system also produces a maintenance summary report in text format. This report provides a brief overview of the simulated fleet, including the total number of machines, alert-level distribution, equipment-category information, and machines that require maintenance attention.

SmartMaint AI also allows users to download an individual maintenance report for a selected machine. The report contains the following information:

- Machine ID and equipment category
- Health score and condition status
- Maintenance priority
- Detected problem
- Observed abnormal pattern
- Recommended action
- Maintenance schedule suggestion
- Short AI summary

The individual report uses a concise structure to avoid repeating the same information in several sections. This provides users with a clear summary of the selected machine without requiring them to review every part of the dashboard.

The maintenance schedule section includes the suggested maintenance type, urgency, schedule status, suggested due date, and reason. These values are produced using rule based decision logic based on the machine health score, alert level, equipment category, and observed condition.

The suggested maintenance activity and due date are intended as decision-support information. They should be reviewed by a qualified technician or engineer before maintenance work is carried out because the system does not confirm the actual physical condition of the equipment.

The generated reports are organised in a clear and readable format so that users can understand the machine condition and recommended maintenance action easily. Overall, the report generation function converts the dashboard results into useful files that can be stored, shared, and used as maintenance records or project documentation.

```
SMARTMAINT AI - SYSTEM SUMMARY REPORT
=====================================

Generated On        : 2026-06-26 00:16:33
Last Data Updated   : 2026-06-23 02:11:03

Model Roles
-----------
Core Prediction Model     : Random Forest
Supporting Anomaly Model  : Isolation Forest
Comparison Anomaly Model  : One-Class SVM

Simulation Scope
----------------
The AI4I dataset provides the sensor readings and model predictions.
Because the dataset does not contain real machine IDs or fleet histories,
SmartMaint AI creates 200 simulated machines with controlled maintenance-
condition scenarios for prototype demonstration. Original sensor readings
and model outputs remain unchanged.

System Summary
--------------
Total Machines      : 200
Equipment Categories: 18
Average Simulated Fleet Health Score: 45.80
Average RF Failure Probability: 7.30%

Alert Distribution
------------------
alert_level
Medium Risk    78
High Risk      59
Low Risk       44
Critical       16
Normal          3

Most Critical Machines
----------------------
machine_id equipment_category  health_score alert_level                                                                   recommended_action
   HEX-010     Heat Exchanger           5.0    Critical            Stop or bypass if required and perform urgent cleaning and flow inspection
   PMP-005               Pump           5.9    Critical Stop pump if necessary and urgently inspect suction line, impeller, seals, and bearings
   MOT-007              Motor           6.8    Critical          Stop or isolate the motor if safety risk exists and perform urgent inspection
   AGT-002           Agitator           7.7    Critical        Stop agitator if unsafe and urgently inspect shaft, blades, coupling, and motor
   GRB-007            Gearbox           8.6    Critical                    Stop gearbox operation if necessary and perform urgent inspection
   CMP-017         Compressor           9.5    Critical         Stop or isolate compressor if required and perform urgent technical inspection
   CMP-012         Compressor          10.4    Critical         Stop or isolate compressor if required and perform urgent technical inspection
   PMP-012               Pump          11.3    Critical Stop pump if necessary and urgently inspect suction line, impeller, seals, and bearings
   AGT-001           Agitator          12.2    Critical        Stop agitator if unsafe and urgently inspect shaft, blades, coupling, and motor
   PMP-003               Pump          13.1    Critical Stop pump if necessary and urgently inspect suction line, impeller, seals, and bearings
```

Figure 5.14: Report Download Maintenance Summary

```
SMARTMAINT AI - MACHINE MAINTENANCE REPORT
=========================================

Machine Information
-------------------
Machine ID          : HYD-006
Equipment Category  : Hydraulic System

Health Summary
--------------
Fleet Health Score  : 38.5 / 100
Fleet Condition     : High Risk
Priority            : Urgent

Detected Problem
----------------
High hydraulic system risk

Observed Abnormal Pattern
-------------------------
anomaly detected, high hydraulic load demand, low operating temperature

Recommended Action
------------------
Plan maintenance soon and reduce hydraulic load if possible

Maintenance Schedule Suggestion
-------------------------------
Maintenance Type    : Hydraulic System Pressure/Leak Inspection
Urgency             : Urgent
Suggested Due Date  : 2026-06-27 (Within 1 day)

AI-Assisted Summary
-------------------
The simulated fleet assessment classified this machine as High Risk.
AI prediction and sensor evidence support the recommended maintenance review.
```

Figure 5.15: Generated Machine Maintenance Report

5.9 Dashboard Development

The SmartMaint AI dashboard was developed using Streamlit because it can display Python data, machine learning results, charts, tables, and interactive controls within one local application. The dashboard runs on the user's personal computer and does not require a cloud server.

The dashboard is connected to the AI analysis pipeline through the output files generated by the system. These files contain information such as the final machine results, simulated machine summary, maintenance alert log, model evaluation results, and Random Forest feature importance. After the AI analysis is completed, the dashboard loads the latest files and displays the results in a structured and understandable form.

The application is divided into four main sections: Dashboard, Machine Monitoring, Maintenance Alert Log, and Reports Centre. Users can move between these sections using the navigation controls. Filters are also provided to help users search by equipment category, alert level, priority, or machine ID.

The Dashboard section provides an overall summary of the controlled simulated fleet. It displays KPI cards, alert-level distribution, machine health information, equipment-category summaries, and machines that require maintenance attention. The machine health graph represents the health scores of different simulated machines and should not be interpreted as a real time-series trend.

The Machine Monitoring section displays the 200 simulated machines in a structured table. Users can review information such as machine ID, equipment category, health score, alert level, priority, and maintenance condition. A selected machine can be opened to view its sensor readings, machine learning evidence, detected problem, probable cause, observed abnormal pattern, recommended action, and maintenance schedule suggestion.

The Maintenance Alert Log section displays machines that require observation, inspection, urgent attention, or immediate maintenance. Users can filter the alerts by equipment category, alert level, priority, and schedule condition. This section helps users identify which machines should receive attention first.

The Reports Centre allows users to preview and download individual machine reports, filtered CSV results, and system summary reports. The reports present the machine condition and maintenance information in a clear format for documentation and future reference.

Random Forest is identified as the main supervised machine failure prediction model. Isolation Forest provides supporting anomaly detection evidence, while One-Class SVM is used as an unsupervised comparison model. The dashboard separates the machine learning evidence from the simulated machine health score so that users do not interpret the health score as a direct failure probability.

HTML and CSS were used together with Streamlit to improve the layout and appearance of the dashboard. They were used to customise cards, buttons, text, spacing, borders, tables, and alert indicators. Different colours were applied to make the Normal, Low Risk, Medium Risk, High Risk, and Critical conditions easier to identify.

The dashboard also includes a **Run AI Analysis** function. This allows the user to execute the local analysis pipeline and regenerate the latest output files. After the process is completed, the updated results are loaded and presented in the dashboard.

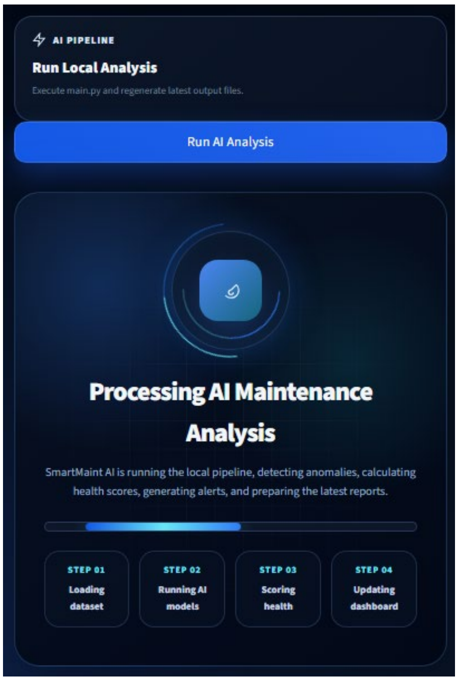


Figure 5.16: Run Local Analysis Button

5.10 Conclusion

This chapter explained the implementation of the SmartMaint AI system based on the requirements and system design discussed in the previous chapters. The system was developed as a local PC-based predictive maintenance application using Python and Streamlit.

The AI4I 2020 Predictive Maintenance dataset was loaded, checked, and preprocessed before being used by the machine learning models. Important features such as machine type, air temperature, process temperature, rotational speed, torque, and tool wear were selected for the analysis. Identification columns were removed, categorical values were encoded, and numerical features were standardised where required.

Three machine learning models were implemented using Scikit-learn. Random Forest was used as the main supervised model for machine failure prediction because it achieved the best overall evaluation results. Isolation Forest was used to provide supporting anomaly detection evidence, while One-Class SVM was used as an unsupervised comparison model.

The original sensor readings and model outputs were kept as AI evidence. Since the dataset does not contain complete machine histories or a real industrial fleet, the system also generated a controlled simulated fleet of 200 machines from 18 equipment categories. This simulation was used to demonstrate machine-level health scores, alert levels, maintenance recommendations, alert logging, scheduling, and report generation.

The system classified machine conditions into Normal, Low Risk, Medium Risk, High Risk, and Critical. It also generated practical maintenance information, including the detected problem, probable cause, observed abnormal pattern, recommended action, maintenance priority, and maintenance schedule suggestion.

The Streamlit dashboard was implemented to present the system results in a simple and interactive form. Users can view the overall machine condition, monitor individual machines, review maintenance alerts, inspect sensor readings, apply filters, and download machine or system summary reports.

CHAPTER 6: TESTING / EVALUATION

6.1 Introduction

This chapter presents the testing and evaluation of the SmartMaint AI system. The purpose of this chapter is to check whether the developed system works according to the requirements and objectives discussed in the previous chapters. The testing covers both the machine learning models and the main functions of the application.

Functional testing was carried out to check important system functions such as loading the dataset, preprocessing the data, running the AI analysis, displaying machine results, filtering machines, viewing selected machine details, generating maintenance information, displaying the Maintenance Alert Log, and downloading reports. Each function was tested by comparing the expected result with the actual result produced by the system.

The machine learning evaluation was conducted to measure the performance of Random Forest, Isolation Forest, and One-Class SVM. The models were evaluated using accuracy, precision, recall, F1-score, ROC-AUC, PR-AUC, and confusion matrix results.

Random Forest was selected as the main supervised machine failure prediction model because it achieved the best overall evaluation results. Isolation Forest was used as a supporting unsupervised anomaly detection model, while One-Class SVM was used as an unsupervised comparison model.

This chapter also evaluates the controlled simulated fleet used for prototype demonstration. The evaluation includes the machine health scores, alert-level distribution, equipment-category results, maintenance recommendations, schedule suggestions, alert logs, dashboard outputs, and generated reports.

The final simulated machine conditions are used to test the machine-level monitoring and maintenance decision-support functions. They should not be interpreted as direct Random Forest failure probabilities or as health scores produced only from Isolation Forest anomaly results.

The evaluation results are discussed to explain the strengths and limitations of SmartMaint AI. This helps determine whether the system can perform machine failure prediction, provide supporting anomaly evidence, present machine conditions clearly, and generate useful information for maintenance decision-making.

6.2 Testing Approach

The SmartMaint AI system was tested using two main approaches, which were functional testing and machine learning model evaluation. These approaches were used to check whether the system functions correctly and whether the machine learning models produce useful prediction results.

Functional testing was carried out on the main system features. This included loading the dataset, preprocessing the data, running the AI analysis, displaying dashboard results, filtering machine records, viewing machine details, generating maintenance alerts, and downloading reports. Each function was tested by comparing the expected result with the actual result shown by the system.

Machine learning model evaluation was carried out using the testing portion of the AI4I 2020 Predictive Maintenance dataset. The dataset was divided into training and testing sets using a stratified split. This helped keep the distribution of normal and failure records similar in both sets because the dataset contains fewer failure records than normal records.

Isolation Forest and One-Class SVM were trained using normal training records so that the models could learn the normal machine operating pattern. Random Forest was trained using both the machine operation features and the machine failure label. The models were then tested using the same testing set to make the comparison fair.

The evaluation results were measured using accuracy, precision, recall, F1-score, ROC-AUC, PR-AUC, and confusion matrix values. These measurements were selected because the dataset is imbalanced and accuracy alone may not clearly show how well the models detect machine failures. The testing results were stored in output files and later used for comparison and discussion.

6.3 Functional Testing

Functional testing was conducted to ensure that the main features of SmartMaint AI operate correctly. A black box testing approach was used, where each function was tested through the user interface without examining its internal source code during the test. The expected result was compared with the actual result produced by the system.

The testing covered the AI analysis process, dashboard display, machine monitoring, filtering functions, maintenance alert log, and report generation. Table 6.1 presents the functional test cases and their results.

Table 6.1: Functional Testing Results of SmartMaint AI

Test ID	Function Tested	Test Procedure	Expected Result	Actual Result	Status
FT01	Launch SmartMaint AI	Run the Streamlit application on the local computer	The application should open without an error and display the main dashboard	The application opened successfully and displayed the dashboard	Passed
FT02	Run AI Analysis	Click the Run AI Analysis button	The system should execute data preprocessing and machine learning analysis	The AI pipeline was executed and the output files were generated successfully	Passed
FT03	Load Machine Results	Open the dashboard after completing the AI analysis	The system should load the latest machine analysis results	The latest machine results were loaded and displayed correctly	Passed

FT04	Display Dashboard Summary	Open the Dashboard section	The system should display health summaries, alert statistics, charts, and recent machine records	The summary cards, charts, and recent records were displayed correctly	Passed
FT05	Filter Machine Records	Select an equipment category, alert level, or machine ID	The displayed results should change according to the selected filter	The machine records and dashboard information were filtered correctly	Passed
FT06	View Machine Details	Select a machine from the Machine Monitoring section	The system should display the selected machine's sensor readings, health score, alert level, and maintenance information	The complete machine information was displayed correctly	Passed
FT07	Display Maintenance Alert Log	Open the Maintenance Alert Log section	The system should display active alerts with machine information, priority, problem, and recommended action	The maintenance alerts and action summaries were displayed correctly	Passed
FT08	Download Filtered CSV Report	Apply filters and click the CSV report download button	The system should generate a CSV file containing the filtered machine records	The filtered CSV report was generated and downloaded successfully	Passed

FT09	Download Maintenance Summary	Click the maintenance summary download button	The system should generate a readable maintenance summary file	The maintenance summary was generated and downloaded successfully	Passed
FT10	Generate Individual Machine Report	Open a selected machine and download its maintenance report	The report should contain the machine information, health score, problem, abnormal pattern, recommended action, and maintenance schedule	The individual machine maintenance report was generated correctly	Passed
FT11	Navigate Between System Sections	Use the sidebar to move between Dashboard, Machine Monitoring, Maintenance Alert Log, and Reports	The selected section should open correctly	All system sections could be accessed without navigation errors	Passed

Based on the functional testing results, all major functions of SmartMaint AI operated according to the expected requirements. The system was able to execute the AI analysis, display the machine results, generate maintenance information, and provide downloadable reports without major functional errors.

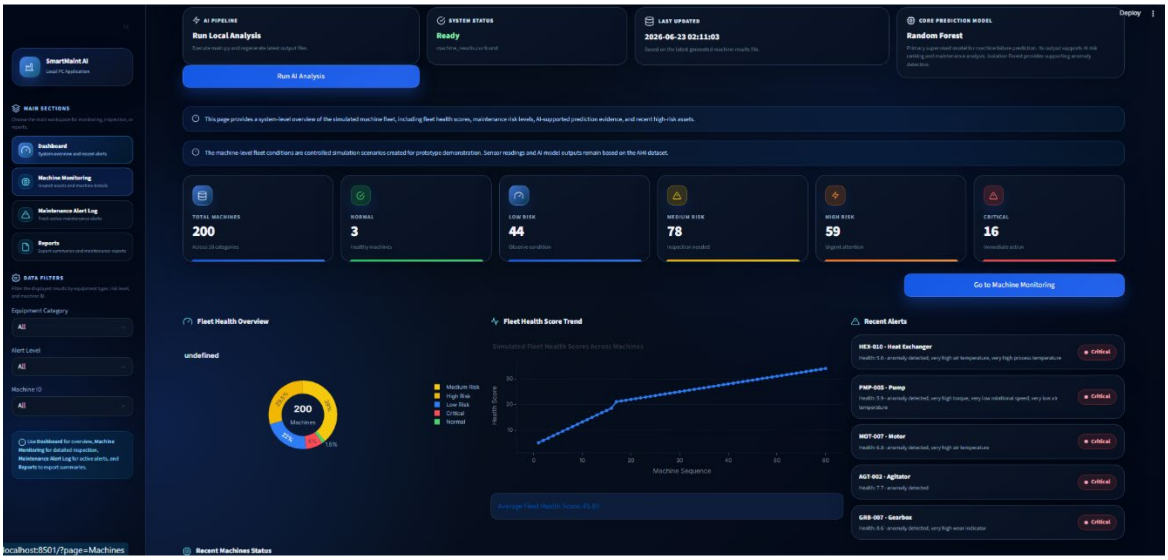


Figure 6.1: Successful Functional Testing of the SmartMaint AI Dashboard

6.4 Machine Learning Model Evaluation

The three machine learning models were evaluated using 2,000 testing records from the AI4I 2020 Predictive Maintenance dataset. The evaluated models were Random Forest, Isolation Forest, and One-Class SVM. The testing records were separate from the records used to train the Random Forest model.

Several performance metrics were used because the dataset contains many normal records and a much smaller number of machine failure records. This class imbalance means that accuracy alone is not enough to explain the performance of a model.

Accuracy measures the overall percentage of correct predictions. Precision measures how many records predicted as failures were actual failures. Recall measures how many actual failure records were successfully detected. The F1-score provides a balance between precision and recall. ROC-AUC measures how well a model separates failure records from normal records across different thresholds, while PR-AUC focuses more strongly on the model's performance for the less common failure class.

Table 6.2: Machine Learning Model Evaluation Results

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	PR-AUC
Isolation Forest	94.55%	17.46%	16.18%	16.79%	80.64%	11.00%
One-Class SVM	94.50%	27.17%	36.76%	31.25%	77.29%	27.33%
Random Forest	98.45%	84.91%	66.18%	74.38%	97.29%	79.67%

Random Forest achieved the strongest overall performance. It obtained an accuracy of 98.45%, precision of 84.91%, recall of 66.18%, and F1-score of 74.38%. It also achieved the highest ROC-AUC and PR-AUC values. These results show that Random Forest was more effective at predicting machine failure than the other two models.

Random Forest achieved better results because it is a supervised model trained using the known Machine failure labels. It learns the relationship between the machine operating features and the actual failure output. Based on these results, Random Forest was selected as the main machine failure prediction model in SmartMaint AI.

One-Class SVM achieved a recall of 36.76%, which was higher than the recall of Isolation Forest. This means that it detected more of the actual failure records. However, its lower precision shows that it also classified more normal records as abnormal. Its F1-score of 31.25% was better than the Isolation Forest result but remained lower than Random Forest.

Isolation Forest achieved an accuracy of 94.55% and a ROC-AUC value of 80.64%. However, its precision, recall, F1-score, and PR-AUC were lower than the other models. Therefore, Isolation Forest was not selected as the main prediction model. It was retained as a supporting unsupervised anomaly detection model because it can identify unusual operating patterns without using the Machine failure label during training.

The evaluation results of the supervised and unsupervised models should be interpreted according to their different purposes. Random Forest directly predicts the labelled machine failure target, while Isolation Forest and One-Class SVM identify records that differ from the learned normal operating pattern. An unusual record does not always represent an actual machine failure.

The results also confirm that accuracy should not be considered alone. Because most testing records represent normal operation, a model can achieve high accuracy by correctly identifying normal records while still missing many failures. For this reason, precision, recall, F1-score, ROC-AUC, and PR-AUC were also considered when selecting the main model.

	A	B	C	D	E	F	G
1	model	accuracy	precision	recall	f1_score	roc_auc	pr_auc
2	Isolation Forest	0.9455	0.17460317	0.161764706	0.16793893	0.806395384	0.11002077
3	One-Class SVM	0.945	0.27173913	0.367647059	0.3125	0.772850445	0.27328789
4	Random Forest	0.9845	0.8490566	0.661764706	0.74380165	0.972898398	0.79666537

Figure 6.2: Evaluation Results of the Machine Learning Models

6.5 Model Comparison

The three machine learning models were compared to understand their strengths, limitations, and roles in the SmartMaint AI system. Although the models were tested using the same dataset, they use different learning approaches and were developed for different purposes. Therefore, the comparison was based on several performance measures and not only on accuracy.

Random Forest achieved the best overall performance among the three models. It obtained the highest accuracy, precision, recall, F1-score, ROC-AUC, and PR-AUC. This model performed well because it was trained using labelled machine failure data. It learned the relationship between the machine operating features and the known failure output.

Based on the evaluation results, Random Forest was selected as the main machine failure prediction model in SmartMaint AI. It provides the predicted failure class and failure probability used as the main machine learning evidence in the system.

One-Class SVM achieved better recall and F1-score than Isolation Forest. This means that it detected more actual failure records during testing. However, its lower precision shows that it also produced more false-positive predictions. One-Class SVM was therefore kept as an unsupervised comparison model to show how another anomaly detection method performs on the same machine operating data.

Isolation Forest produced lower precision, recall, F1-score, and PR-AUC compared with the other models. For this reason, it was not selected as the main machine failure prediction model. However, it was retained as a supporting unsupervised anomaly detection model because it can identify unusual operating patterns without using the Machine failure label during training.

The comparison also shows that the supervised and unsupervised models should not be interpreted in exactly the same way. Random Forest directly predicts the labelled machine failure target, while Isolation Forest and One-Class SVM detect records that are different from the learned normal pattern. An unusual record may indicate abnormal behaviour, but it does not always mean that an actual machine failure has occurred.

Table 6.3: Comparison of the Machine Learning Models

Model	**Learning Type**	**Main Strength**	**Limitation**	**Role in SmartMaint AI**
Random Forest	Supervised	Achieved the highest overall performance and provides failure probability	Requires labelled machine failure data	Main machine failure prediction model
Isolation Forest	Unsupervised	Detects unusual operating patterns without using failure labels	Lower failure classification performance	Supporting anomaly detection model
One-Class SVM	Unsupervised	Detected more actual failure records than Isolation Forest	Produced more false-positive results and is sensitive to scaling and parameters	Unsupervised comparison model

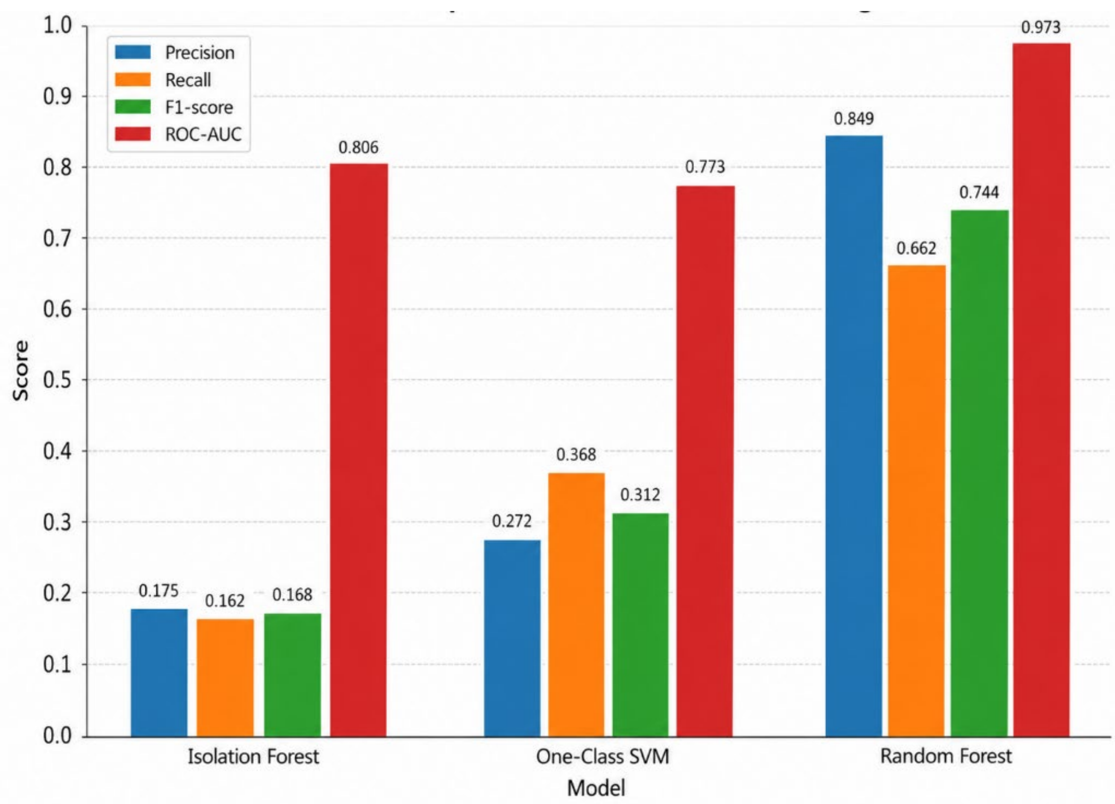


Figure 6.3: Performance Comparison of the Machine Learning Models

6.6 Random Forest Feature Importance

Random Forest feature importance was analysed to identify which input features had the greatest influence on machine failure classification. The results were obtained from the trained Random Forest model and saved in the random_forest_feature_importance.csv file.

Torque was the most influential feature, with an importance value of 33.29%. Rotational speed was the second most important feature at 28.02%, followed by tool wear at 21.01%. Air temperature contributed 10.17%, while process temperature contributed 6.18%. Machine type had the lowest importance value at 1.32%.

Table 6.4: Random Forest Feature Importance Results

Feature	Importance
Torque	33.29%
Rotational Speed	28.02%
Tool Wear	21.01%
Air Temperature	10.17%
Process Temperature	6.18%
Machine Type	1.32%

The results show that mechanical operating conditions, particularly torque and rotational speed, had a strong influence on machine failure classification. Tool wear was also important because increasing tool usage may affect machine performance and failure risk. Temperature features had a smaller but still meaningful contribution.

Machine type had the lowest importance value, which indicates that the operating sensor readings had a greater influence on the Random Forest predictions than the machine type category. However, feature importance only shows the contribution of each input to the model prediction and does not prove that a feature directly caused the machine failure.

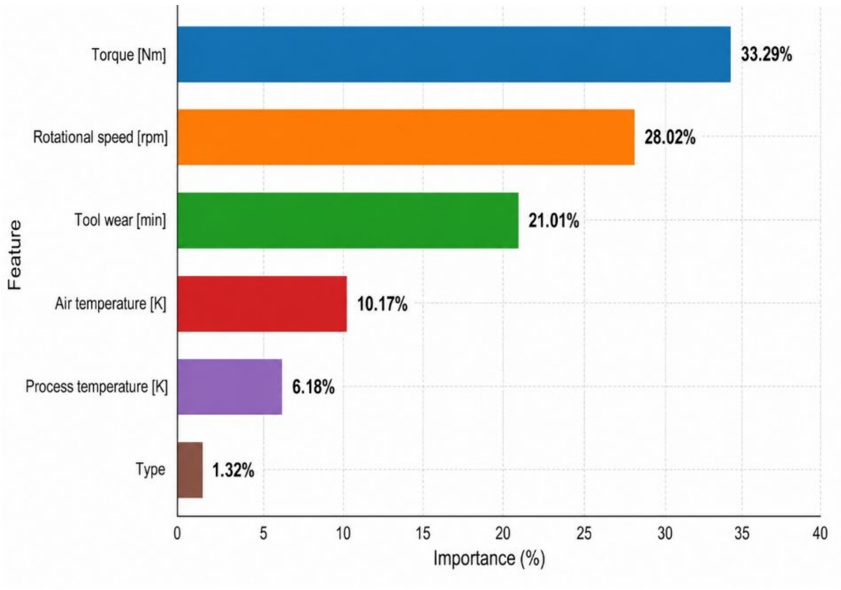


Figure 6.4: Random Forest Feature Importance Results

6.7 Health Score and Alert Level Evaluation

The health score and alert-level outputs were evaluated to check whether SmartMaint AI could present machine conditions in a clear and understandable form. Each machine in the controlled simulated fleet was assigned a health score ranging from 0 to 100. A higher score represents a healthier machine condition, while a lower score represents a higher level of maintenance attention.

The final machine-level health scores were generated as part of the controlled simulation used for prototype demonstration. They were not calculated directly from the Random Forest failure probability or the Isolation Forest anomaly score. The original sensor readings and machine learning outputs were kept separately as AI evidence.

Based on the simulated health score, each machine was classified into one of five alert levels:

- **Normal:** 80 to 100
- **Low Risk:** 60 to below 80
- **Medium Risk:** 40 to below 60
- **High Risk:** 20 to below 40
- **Critical:** Below 20

The system also assigned a suitable maintenance priority. Normal machines were given the **Stable** priority, Low Risk machines were assigned **Observe**, Medium Risk machines were assigned **Inspect**, High Risk machines received **Urgent**, and Critical machines received **Immediate** priority.

Table 6.5: Sample and Example Health Score and Alert-Level Results

Machine ID	Equipment Category	Health Score	Alert Level	Priority
MOT-004	Motor	80.19	Normal	Stable
MOT-001	Motor	71.69	Low Risk	Observe
MOT-011	Motor	42.15	Medium Risk	Inspect
CMP-005	Compressor	29.56	High Risk	Urgent
CMP-002	Compressor	9.80	Critical	Immediate

The sample results show that the health-score ranges were applied correctly. For example, MOT-004 received a health score of 80.19 and was classified as Normal. MOT-001 received a score of 71.69 and was classified as Low Risk.

MOT-011 received a health score of 42.15 and was classified as Medium Risk with Inspect priority. CMP-005 received a lower score of 29.56 and was classified as High Risk with Urgent priority. CMP-002 received the lowest score of 9.80 and was correctly classified as Critical with Immediate priority.

The results show a clear relationship between the simulated health score, alert level, and maintenance priority. As the health score decreases, the level of maintenance attention increases. This makes it easier for users to identify which machines can continue under routine monitoring and which machines require inspection, urgent maintenance, or immediate action.

This distribution was intentionally controlled for prototype demonstration. It allowed the system to test all dashboard cards, filters, machine-monitoring functions, maintenance recommendations, alert logs, scheduling functions, and reports.

6.8 Maintenance Output Evaluation

The maintenance outputs were evaluated to check whether SmartMaint AI could convert the simulated machine conditions and available machine evidence into clear and useful maintenance information. The evaluation focused on the detected problem, probable cause, observed abnormal pattern, recommended action, maintenance priority, and maintenance schedule suggestion.

The maintenance outputs were generated using rule-based decision logic. The rules considered the equipment category, health score, alert level, maintenance priority, and available sensor patterns. This allowed the system to provide different maintenance information for different machine conditions instead of producing the same recommendation for every machine.

Machines from all five alert levels were reviewed during the evaluation. Normal machines were given routine monitoring suggestions, while Low Risk machines received observation-based actions. Medium Risk machines were given inspection recommendations, High Risk machines received urgent maintenance suggestions, and Critical machines were assigned immediate actions.

The maintenance priorities were also checked against the alert levels. Normal machines were assigned Stable priority, Low Risk machines were assigned Observe, Medium Risk machines were assigned Inspect, High Risk machines received Urgent priority, and Critical machines received Immediate priority. The results showed that the assigned priorities were consistent with the defined alert-level rules.

The detected problems and probable causes were also reviewed according to the equipment category. For example, a pump may receive possible causes related to cavitation, bearing wear, or unstable hydraulic load, while a motor may receive possible causes related to overheating, excessive load, or rotational instability. This equipment-specific approach made the generated explanations more relevant to each machine type.

The observed abnormal pattern was generated using the available sensor readings and supporting machine learning evidence. Unusual air temperature, process temperature, rotational speed, torque, or tool wear values were included in the explanation when they moved away from the expected operating pattern. However, these patterns were treated as supporting evidence and not as confirmed physical faults.

The recommended actions were evaluated to check whether they matched the machine condition. Machines with lower-risk conditions received monitoring or basic inspection suggestions, while High Risk and Critical machines received earlier and more urgent maintenance actions. Examples included inspecting bearings, checking cooling conditions, reducing operating load, examining lubrication, and stopping the machine for immediate inspection when necessary.

The maintenance schedule suggestions were evaluated by checking the maintenance type, urgency, schedule status, suggested due date, and reason. Higher-risk machines received earlier due dates than lower-risk machines. This showed that the scheduling rules responded correctly to changes in health score and alert level.

The generated outputs were also checked in the Maintenance Alert Log, selected machine details, and downloaded reports. The same machine condition produced consistent maintenance information across these system sections.

6.9 Dashboard and Report Testing

The dashboard and report functions were tested to ensure that users could view the analysis results clearly and download the generated maintenance information. The testing covered navigation, filtering, chart display, machine details, maintenance alerts, and report downloads.

The sidebar navigation was tested by opening the Dashboard, Machine Monitoring, Maintenance Alert Log, and Reports sections. Each section opened successfully and displayed the correct content. The equipment category, alert level, and machine ID filters were also tested. The displayed records changed according to the selected filter values.

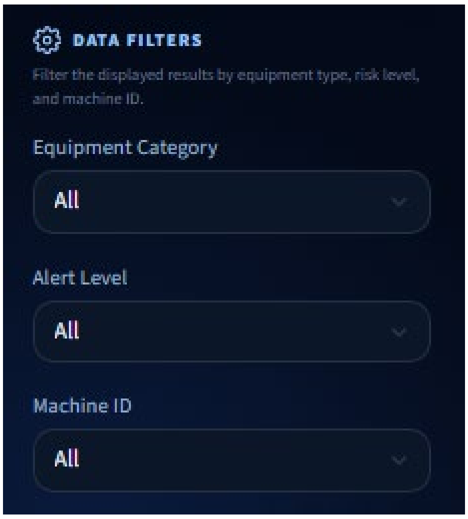


Figure 6.5: Dashboard Filtering

The dashboard charts and summary cards were checked after the AI analysis was completed. The system successfully displayed the total number of machines, alert level distribution, health score trend, recent alerts, and selected machine details. This confirmed that the dashboard could load and present the generated output files correctly.

The report functions were tested by downloading the filtered CSV system report and the text maintenance summary. The downloaded CSV file contained the machine records based on the active filters. The maintenance summary contained the main condition and maintenance information in a readable text format.

machine_id	equipment_category	Type	Air tempe	Process te	Rotationa	Torque [N	Tool wear	Machine f	TWF	HDF	PWF	OSF	RNF
HYD-001	Hydraulic System	2	300.4	311.8	2496	13.5	5	0	0	0	0	0	0
HEX-009	Heat Exchanger	1	303.7	312	2663	11.4	71	1	0	0	1	0	0
FAN-012	Fan	0	295.7	306.2	2270	14.6	149	1	0	0	1	0	0
FAN-004	Fan	0	296.9	307.5	2721	9.3	18	1	0	0	1	0	0
GRB-004	Gearbox	1	302.7	311.6	2709	9.7	2	1	0	0	1	0	0
HYD-002	Hydraulic System	1	304.2	313.7	2113	18.4	139	0	0	0	0	0	0
CHL-009	Chiller	0	304.3	313.5	2643	12.7	21	0	0	0	0	0	0
CMP-002	Compressor	0	296.4	307.4	2833	5.6	213	1	0	0	1	0	0
AGT-001	Agitator	0	303.8	313.1	2497	13	5	1	0	0	1	0	0
PMP-011	Pump	2	297.2	308.5	2440	14	166	0	0	0	0	0	0
GEN-007	Generator	2	300.4	309.9	2563	12.8	81	1	0	0	1	0	0
GEN-006	Generator	2	298.3	308.1	2636	12.8	84	0	0	0	0	0	0
MOT-010	Motor	2	297	308.1	2540	13.3	98	0	0	0	0	0	0
GEN-001	Generator	1	303.4	312.6	2706	9.8	65	1	0	0	1	0	0
CMP-012	Compressor	1	301.5	310.4	2639	12	21	1	0	0	1	0	0
VPM-004	Vacuum Pump	1	303.6	312.8	2659	11.4	26	1	0	0	1	0	0
FAN-001	Fan	1	301.8	310.1	2372	13.9	205	1	0	0	1	0	0
FAN-007	Fan	2	303.6	312.4	1460	53.7	214	0	0	0	0	0	0
CMP-015	Compressor	0	300.4	310.5	2424	14.5	12	0	0	0	0	0	0
CLT-006	Cooling Tower	0	301.5	310.9	2760	8	15	1	0	0	1	0	0
CMP-007	Compressor	1	299.1	308.6	2329	14.6	95	0	0	0	0	0	0

Figure 6.6: Filtering and Machine Result Testing

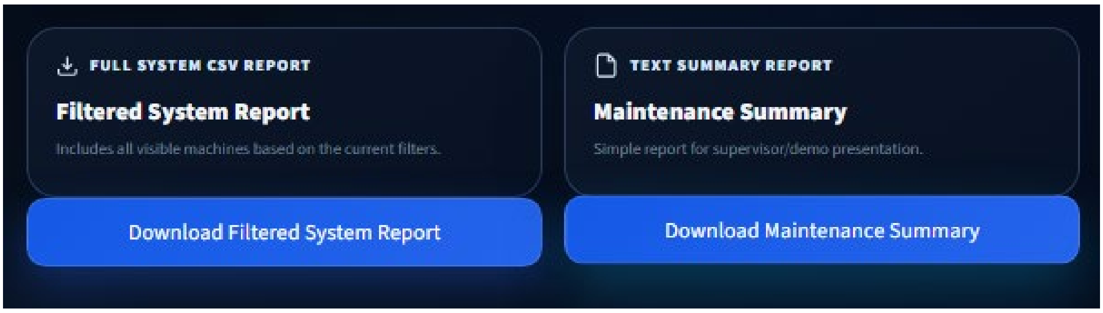


Figure 6.7: Maintenance Report Generation

The individual machine report was also tested using a selected machine. The generated report included the machine information, health score, condition status, priority, detected problem, observed abnormal pattern, recommended action, maintenance schedule suggestion, and AI summary. The reports were generated and downloaded without major errors. the dashboard and report testing showed that SmartMaint AI successfully presented the machine analysis results and allowed users to save the information for maintenance planning and documentation.

6.10 Discussion

The testing and evaluation results showed that SmartMaint AI was able to perform the main functions required for the project. The system successfully processed the AI4I dataset, predicted machine failure, generated supporting anomaly results, prepared the controlled simulated fleet, assigned health scores and alert levels, produced maintenance recommendations, displayed the results through the dashboard, and generated downloadable reports.

The machine learning evaluation showed that Random Forest achieved the strongest overall performance. It obtained the highest accuracy, precision, recall, F1-score, ROC-AUC, and PR-AUC among the three evaluated models. This was mainly because Random Forest is a supervised model trained using known Machine failure labels. Based on these results, Random Forest was selected as the main machine failure prediction model in SmartMaint AI.

One-Class SVM detected more actual failure records than Isolation Forest and achieved higher recall and F1-score. However, it also produced more false-positive predictions. It was therefore retained as an unsupervised comparison model.

Isolation Forest achieved lower precision, recall, F1-score, and PR-AUC than the other models. Therefore, it was not selected as the main prediction model. However, it remained useful as a supporting unsupervised anomaly detection model because it can identify unusual machine operating patterns without using failure labels during training.

The evaluation also showed that accuracy alone was not sufficient for comparing the models. Most records in the AI4I dataset represent normal machine conditions, while only a small number represent machine failures. A model may therefore achieve high accuracy by correctly predicting many normal records while still missing actual failures. For this reason, precision, recall, F1-score, ROC-AUC, and PR-AUC were also considered when selecting the main model.

The controlled simulated fleet allowed the machine-level monitoring functions to be tested using 200 machines from 18 equipment categories. Each simulated machine was assigned a health score and classified as Normal, Low Risk, Medium Risk, High Risk, or Critical. The final distribution contained 3 Normal, 44 Low Risk, 78 Medium Risk, 59 High Risk, and 16 Critical machines.

The simulated health scores and alert levels made the machine conditions easier to understand. However, these values were prepared for prototype demonstration and were not calculated directly from the Random Forest failure probability or the Isolation Forest anomaly score. The original sensor values and model outputs were retained separately as machine learning evidence.

The maintenance priority and rule-based recommendations helped users identify which machines required routine monitoring, observation, inspection, urgent maintenance, or immediate action. The generated information included the detected problem, probable cause, observed abnormal pattern, recommended action, and maintenance schedule suggestion.

The dashboard and reporting functions improved the usability of the system. Users could filter machine records, review selected machine details, view maintenance alerts, examine operating data, and download individual or system-level reports. The same machine condition and maintenance information were presented consistently across the dashboard, alert log, and generated reports.

However, the project has several limitations. The AI4I 2020 Predictive Maintenance dataset is synthetic and may not represent every condition found in real industrial environments. It also does not provide complete historical time-series data for individual rotating machines. The current system uses stored dataset records and a controlled simulated fleet instead of live sensor readings from physical equipment.

The maintenance recommendations and schedule suggestions are intended only as decision-support information. They do not confirm the actual physical condition or cause of a machine problem and should not replace the judgement of qualified technicians or engineers.

Future improvements should include testing the system using real industrial datasets, connecting it to live sensors, adding sensor measurements such as vibration, electrical current, and pressure, and evaluating the models using longer machine operating histories. The health-score method and maintenance rules could also be calibrated using real equipment information and technician feedback.

6.11 Conclusion

This chapter presented the testing and evaluation of the SmartMaint AI system. Functional testing confirmed that the main system features operated successfully, including running the AI analysis, displaying dashboard results, filtering machine records, viewing machine details, generating maintenance information, and downloading reports.

The machine learning evaluation compared Random Forest, Isolation Forest, and One-Class SVM. Random Forest achieved the best overall performance and was selected as the main machine failure prediction model. Isolation Forest was retained as a supporting anomaly detection model, while One-Class SVM was used as an unsupervised comparison model.

The evaluation also confirmed that the controlled simulated fleet could present machine conditions through health scores, alert levels, priorities, maintenance recommendations, and schedule suggestions. The dashboard and report functions displayed and exported this information clearly.

CHAPTER 7: CONCLUSION

7.1 Introduction

This chapter presents the overall conclusion of the SmartMaint AI project and summarises the main outcomes achieved during the system development. The project was developed to demonstrate how artificial intelligence can support predictive maintenance by analysing machine operating data, detecting abnormal conditions, and presenting useful maintenance information to users.

The chapter reviews the achievement of the project objectives, the main contributions of the developed system, and the limitations identified during implementation and testing. It also provides recommendations for future improvements, including the use of real industrial data, live sensor integration, and further enhancement of the machine learning models. this chapter evaluates the final outcome of SmartMaint AI and explains how the project can be improved and extended in future work.

7.2 Project Summary

SmartMaint AI was developed as a local PC-based predictive maintenance system for rotating equipment. The system uses machine operating data to predict machine failure, identify unusual behaviour, and provide maintenance-related information through an interactive Streamlit dashboard.

The AI4I 2020 Predictive Maintenance dataset was used to develop and evaluate the system. The dataset contains machine operating features such as air temperature, process temperature, rotational speed, torque, tool wear, and machine type. The data was cleaned, transformed, encoded, and standardised where required before being used by the machine learning models.

Random Forest was selected as the main supervised machine failure prediction model because it achieved the best overall evaluation results. Isolation Forest was used as a supporting unsupervised anomaly detection model, while One-Class SVM was implemented as an unsupervised comparison model.

The system also generated a controlled simulated fleet of 200 machines from 18 rotating-equipment categories for prototype demonstration. The simulated machine conditions were presented using health scores, alert levels, and maintenance priorities. The system also generated detected problems, probable causes, observed abnormal patterns, recommended actions, and maintenance schedule suggestions using rule-based decision logic.

The results were displayed through the Dashboard, Machine Monitoring, Maintenance Alert Log, and Reports Centre sections. Users could filter machine records, view individual machine details, review maintenance alerts, and download filtered CSV files, individual machine reports, and system summary reports.

The testing results showed that the main system functions operated correctly. Overall, SmartMaint AI achieved the main project purpose of combining machine failure prediction, supporting anomaly detection, machine-condition monitoring, and maintenance decision support in one local application.

7.3 Project Contributions

The SmartMaint AI project provides several contributions to the area of predictive maintenance. The main contribution is the development of a complete local PC-based application that combines machine learning analysis, machine-condition monitoring, maintenance recommendations, alert management, visualisation, and report generation in one system.

The project demonstrates the use of supervised and unsupervised machine learning techniques for analysing machine operating data. Random Forest was selected as the main supervised machine failure prediction model because it achieved the best overall evaluation results. Isolation Forest was used to provide supporting anomaly detection evidence, while One-Class SVM was used as an unsupervised comparison model.

Another contribution is the conversion of technical machine learning evidence into understandable maintenance information. Instead of displaying only predicted classes, failure probabilities, anomaly labels, and anomaly scores, the system provides health scores, alert levels, maintenance priorities, detected problems, probable causes, observed abnormal patterns, recommended actions, and maintenance schedule suggestions.

The project also provides a controlled simulated fleet of 200 machines from 18 rotating-equipment categories. This simulation allows the system to demonstrate machine monitoring, risk classification, maintenance alert generation, scheduling, and reporting for different machine conditions.

The Streamlit dashboard improves the presentation and accessibility of the predictive maintenance results. Users can monitor machine conditions, filter records, review maintenance alerts, examine selected machine details, and download machine and system reports through a clear interface.

In addition, the project provides an educational example of how Python, machine learning, data processing, visualisation, and rule-based decision support can be integrated into a practical artificial intelligence application. The developed system may also serve as a foundation for future predictive maintenance projects using real industrial datasets and live sensor readings.

7.4 Project Limitations

Although SmartMaint AI achieved its main objectives, several limitations were identified during development and testing.

The first limitation is the use of the AI4I 2020 Predictive Maintenance dataset. The dataset is synthetic and may not fully represent sensor noise, environmental changes, measurement errors, and operating conditions found in real industrial environments. Therefore, the model performance may be different when tested using real machines.

The system also analyses stored dataset records instead of receiving live sensor data. It does not currently provide continuous real-time monitoring or automatic updates from physical equipment. Users must run the AI analysis to generate updated results.

Another limitation is that the 200-machine fleet, health scores, and alert-level distribution were created through a controlled simulation for prototype demonstration. These outputs do not represent real machine histories and were not calculated directly from Random Forest failure probabilities or Isolation Forest anomaly scores.

The available features are limited to machine type, air temperature, process temperature, rotational speed, torque, and tool wear. Other useful measurements, such as vibration, pressure, electrical current, sound, oil condition, and humidity, are not included.

The maintenance explanations and schedule suggestions are generated using rule-based decision-support logic. They provide useful guidance but cannot confirm the actual physical cause of a machine problem and should not replace inspection by qualified technicians or engineers.

Finally, the application is designed to run locally on one computer. It does not include cloud access, user accounts, multiple-user management, live sensor integration, or direct connection with industrial maintenance systems.

7.5 Future Work

Several improvements can be carried out in the future to increase the accuracy, usefulness, and practical value of SmartMaint AI. One important improvement is to test the system using real industrial machine data. Real datasets may contain sensor noise, missing values, changing operating conditions, and different failure patterns. This would provide a more realistic evaluation of the system.

The system could also be connected to live sensors for continuous machine monitoring. Sensor readings could be collected and analysed automatically, allowing the system to generate alerts when unusual or high-risk conditions are detected.

Additional sensor measurements, such as vibration, electrical current, pressure, sound, humidity, and oil condition, could also be included. In particular, adding current and pressure data would help the system identify a wider range of machine conditions that are not available in the current dataset.

Future versions may use additional datasets and develop separate models for different equipment categories, such as motors, pumps, fans, compressors, and gearboxes. This could make the failure predictions and maintenance recommendations more specific to each equipment type.

The controlled health-score method, alert thresholds, and maintenance rules should also be calibrated using real machine histories, operating hours, maintenance records, and feedback from qualified technicians. This would improve the reliability of the machine-condition and scheduling outputs.

Other improvements may include model parameter tuning, automatic retraining, time-series analysis, cloud deployment, database storage, user accounts, notifications, and integration with maintenance management systems. These improvements could transform SmartMaint AI from a local prototype into a more complete industrial predictive maintenance platform.

7.6 Conclusion

This chapter presented the overall conclusion of the SmartMaint AI project. The system was successfully developed as a local PC-based predictive maintenance application for rotating equipment. It combines data preprocessing, machine learning, machine-condition monitoring, maintenance decision support, alert management, dashboard visualisation, and report generation.

Random Forest was selected as the main supervised machine failure prediction model because it achieved the strongest overall evaluation results. Isolation Forest was used as a supporting anomaly detection model, while One-Class SVM was used as an unsupervised comparison model.

The system also used a controlled simulated fleet to demonstrate health scores, alert levels, maintenance priorities, detected problems, recommended actions, and maintenance schedule suggestions. Testing confirmed that the main functions operated correctly and that the dashboard presented the results clearly.

Although the current project uses synthetic data, stored records, and simulated machine conditions, it provides a useful foundation for future development. Real industrial datasets, live sensors, additional measurements, and maintenance history could improve the system further.

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